

The lexical pragmatics of reflexive marking

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To appear in *Language*

June 7, 2025

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Abstract. In French, a subclass of anticausative verbs is optionally marked with the clitic *se*, traditionally considered a reflexive marker. We show that this optionality does not consist of free variation. Rather, the presence or absence of *se* is influenced by lexical-pragmatic considerations: while by default, both variants are equally acceptable, in the context of a human subject, cooperative speakers strongly prefer the variant that in certain cases avoids and in other cases maintains ambiguity with the semantically reflexive interpretation which arises in parallel with the intended (anticausative) interpretation. Understanding these preferences requires taking into account the agent bias, the tendency to interpret human nouns in subject position as agents whenever is possible, and the multifunctionality of *se*, which is used both to form (non-agentive) anticausative predicates as well as (agentive) semantically reflexive ones. Depending on whether the alternative reflexive interpretation is in line with shared assumptions about the event, the preference for the presence vs. absence of *se* is predicted. The interaction between the choice of form by the cooperative language user and individual verb subclasses is an example of what we call lexical-pragmatic effects.*

Keywords: causative alternation, reflexive, French, change-of-state verbs, lexical pragmatics, agentivity, ambiguity

1. INTRODUCTION. Change-of-state verbs with a transitive and an intransitive use, such as English *break* and *open*, are said to undergo the CAUSATIVE/ANTICAUSATIVE ALTERNATION. In their intransitive use, they describe a change-of-state event undergone by the internal argument. In their transitive use, the entity causing the change-of-state is named by the external argument of the (now two-place) predicate. The transitive use is sometimes paraphrased as ‘cause to V[intransitive]’ (Levin & Rappaport Hovav 1995: 79), although finding the exact characterisation across languages raises a range of questions for theories of syntax, morphology and semantics (Haspelmath 1993, Alexiadou et al. 2015, Martin et al. 2025, a.o.).

In French, as in other Romance languages and beyond, verbs undergoing the causative/anticausative alternation are divided into two morphological and three distributional classes, depending on whether their anticausative variant does or does not co-occur with

*For feedback, we thank the audience of the University of Edinburgh, Humboldt-Universität zu Berlin, LAGB 2023 and Geneva University. Many thanks also to Stefano Coretta and Henriëtte de Swart. F. Martin’s research in the first phase of this project was supported by DFG award AL 554/8-1 (Leibniz-Preis 2014) to Artemis Alexiadou and by the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (LeibnizDream, grant agreement No 856421). Thanks to Artemis Alexiadou for providing financial support for the experiments through her Leibniz-Preis 2014. Finally, we are also grateful to our anonymous referees for their constructive and substantial feedback and to John Beavers and Jesse Harris for their editorial work.

the ‘reflexive’ or more accurately NON-ACTIVE clitic *se* (Zribi-Hertz 1982, 1987; Labelle 1992; Schäfer 2008; Heidinger 2010, 2015, Alexiadou et al. 2015, a.o.). With verbs of the first class illustrated in 1, the anticausative variant (AC) is necessarily unmarked, not differing morphologically from its causative counterpart; we call these anticausatives UNMARKED ANTICAUSATIVES and notate them as ‘*−se*’ AC-verbs, because they are incompatible with *se* under the anticausative reading.¹

(1) Unmarked anticausative, *−se* AC

- a. Ana change la couleur du mur.
Ana changes the color of the wall
‘Ana is changing the color of the wall.’
- b. La couleur du mur (#se) change.
the color of the wall SE changes
‘The color of the wall is changing.’

With verbs of the second class, illustrated in 2, the anticausative variant is obligatorily marked with *se*. We call these MARKED ANTICAUSATIVES and notate them as ‘*+se*’ AC-verbs.

(2) Marked anticausative, *+se* AC

- a. Le temps qui passe amoche tout.
the time that passes damages everything
‘The passage of time damages everything.’
- b. Tout *(s’)amoche avec le temps qui passe.
everything SE damages with the time that passes
‘Everything gets damaged with the passage of time.’

The third class is illustrated in 3. Since the anticausative variants of these verbs allow both markings, we label them OPTIONALLY MARKED ANTICAUSATIVES, ‘*±se*’ AC-verbs.

(3) Optionally marked anticausative, *±se* AC

- a. Gaston casse le vase.
Gaston breaks the vase
‘Gaston is breaking the vase.’

- b. Le vase (se) casse.
 the vase SE breaks
 ‘The vase is breaking.’

A decisive difference between the forms with and without *se* is that the form without *se* has only one parse and interpretation, where the DP’s referent is a theme (and not an agent) of the VP-event, while the corresponding form marked with *se* is formally ambiguous (Ruwet 1972, Zribi-Hertz 1982, 1987, Martin & Schäfer 2014; see also Bahrt 2021 and literature cited therein on the reflexive-anticausative syncretism cross-linguistically): it can either involve an anticausative verb denoting a one-place predicate of change, whose sole internal argument is associated with the theme role only, or a two-place predicate of caused change that underwent reflexivization, such that the nominative DP is associated with both the theme and agent roles. In principle then, the form marked with *se* in 2b and 3b can be either interpreted as an anticausative verb, or as a semantically reflexive verb.

Many Indo-European languages show a similar distribution, with the qualification that the morphological marker found with a subset of anticausatives can be either a clitic as in French (e.g. all Romance languages), a weak pronoun (e.g. German) or a verbal affix (e.g. Icelandic, Russian, Greek). However, French is special insofar as the set of $\pm se$ AC-verbs as in 3b is rather big in this language compared to other Indo-European languages (e.g. Schäfer 2008, Alexiadou et al. 2015).

In this paper, we investigate whether the choice between the variant with or without *se* in 3b is really free or whether there are semantic or pragmatic factors that enforce the presence or absence of the clitic *se* with $\pm se$ AC-verbs. We argue that while the two variants are generally in free variation, this overall optionality tends to be resolved in specific contexts to either the presence or the absence of *se* due to what we consider lexical pragmatic considerations. While, by default, both variants of the AC are equally acceptable and semantically identical, cooperative speakers following the Gricean conversational maxims (Grice 1975) favor the presence or the absence of *se* in particular contexts, if they, thereby, can avoid unintended inferences on the part of the hearer, by avoiding or maintaining the ambiguity of the *se*-form, which can be interpreted not only as an anticausative verb but also as a semantically reflexive verb.

1.1. THE UNMARKED LIMITED-CONTROL AND THE MARKED IN-CONTROL PREFERENCES WITH HUMANS. Our main empirical contribution consists of three interrelated generalizations, each substantiated by an acceptability rating study presented in Section 3. This contribution rests on the idea that the semantic information conveyed by

so-called ‘in-control’ or ‘limited-control’ verbal morphologies familiar from some work on Tagalog and Salish (Dell 1983, Thompson 1988, Jacobs 2011 a.o) can be lexicalized by alternating verbs in languages like French or English.² What we call IN-CONTROL VERBS denote changes typically under the control of humans involved; see *smash the window*.³ LIMITED-CONTROL VERBS denote changes typically not under the control of humans involved; see *drop the glass*. Other verbs are neutral with respect to the control exerted or not by humans involved in the events in their denotation; see *break the glass*. These distinctions are also operational when alternating verbs are used as anticausatives: we have default assumptions about the degree of control human undergoers exert on changes they endure; compare for instance *to age* (human undergoers typically have little control on ageing) and *to move* (human undergoers typically control their movements).⁴

It turns out that such lexical biases reflecting general assumptions on changes involving human undergoers are very relevant for the choice between the variant with and without *se* of $\pm se$ ACs in the context of a human DP in French. What we see is that while $\pm se$ AC-verbs always allow both the uses with and without *se* in the context of an inanimate undergoer as in 3b, the lexical-semantic sub-classes of $\pm se$ AC-verbs just distinguished tend to enforce or prohibit the appearance of *se*, but only when the sole DP-argument of the anticausative predicate is *human*. With limited-control verbs like (*se*) *rougir* ‘blush/redden’, the variant with *se* of $\pm se$ verbs becomes dispreferred if the nominative DP-argument is human, as in 4a-b. When the DP is non-human, the two variants are in free variation, see 4c. We refer to this first generalization as the UNMARKED LIMITED-CONTROL PREFERENCE (for human arguments). But with in-control verbs like (*se*) *plier* ‘bend, fold’, it is the variant without *se* which becomes dispreferred, and this when the DP-argument is human only, as illustrated in 5a-c. We refer to this second generalization as the MARKED IN-CONTROL PREFERENCE (for human arguments). While the anticausative variants 4b and 5a are clearly dispreferred, they are not ungrammatical, however. Although much less frequent, they do occasionally appear in corpora (see section 3 and Appendix 2 for some naturally occurring examples from native speakers of French).

(4) *Unmarked limited-control preference with $\pm se$ anticausatives*

- a. Jeanne a rougi (sous l’effet des compliments).(limited-control verb)
Jeanne has reddened under the effect of.the compliments
‘Jeanne blushed/reddened (under the effect of the compliments).’
- b. #Jeanne s’est rougie (sous l’effet des compliments).
Jeanne SE is reddened under the effect of.the compliments
‘Jeanne blushed/reddened under the effect of the compliments.’

- c. La peau de Jeanne a rougi/ s'est rougie (sous l'effet du soleil).
 the skin of Jeanne has reddened SE is reddened under the effect of the sun
 'Jeanne's skin reddened (from the sun).'

(5) *Marked in-control preference with $\pm se$ anticausatives*

- a. #Jeanne a plié en deux (de douleur). (in-control verb)
 Jeanne has bent in two from pain
 Intended: 'Jeanne bent over (in pain).'
- b. Jeanne s'est pliée en deux (de douleur).
 Jeanne SE is bent in two from pain
 'Jeanne bent over (in pain).'
- c. La branche a plié/ s'est pliée en deux (sous l'effet du vent).
 the branch has bent SE is bent in two under the effect of the wind
 'The branch bent in two (from the wind).'

In each of the examples in 4 and 5, a PP introduces the causer of the event. This reinforces an anticausative construal of the verb (Schäfer 2009), and weakens the (competing) semantically reflexive reading of the form with *se*, where the DP is also an agent, and not only a theme in the semantic representation.

We propose that the above preferences follow from the interplay of default expectations about the role of humans in the events in the denotation of verbs like in 4 and 5 with the set of semantic parses made available by the grammar for the strings with and without *se*: the strings in 4a and 5a without *se* only have one parse and interpretation (the DP's referent is associated with the role of theme but not the role of agent), while the corresponding strings marked with *se* in 4b and 5b are ambiguous (the DP is associated with the role of theme only, or with both the theme and agent roles). On this view, the unmarked limited control and marked in-control preferences reflect the most cooperative choice between forms, as the alternative form triggers unintended inferences on the part of the hearer. This idea is supported by the fact that these preferences exclusively arise with $\pm se$ AC-verbs, for which there is a choice between forms. The class of $-se$ AC-verbs (i.e. verbs for which the anticausative must be formed without *se*) includes in-control verbs (e.g. *changer de position* 'change in position') and the class of $+se$ AC verbs (for which the anticausative must be marked with *se*) includes limited-control verbs (e.g. *s'affaiblir* 'weaken', *s'anémier* 'become anaemic'). Despite the forced absence or presence of *se* for the anticausative parse, they are completely fine with a human DP.

Therefore, the infelicitous cases observed with verbs cannot be just due to some inherent incompatibility between the presence or absence of *se*-marking and the default meaning of the verb (in-control or limited-control). Otherwise, infelicity should be observed with limited-control/in-control verbs across the three morphological subclasses of ACs. Instead, what we see is that *-se* and *+se* verbs superimpose their limited-control or in-control lexical bias onto whatever form they must get when used as ACs.⁵

1.2. THE MARKED RESPONSIBILITY PREFERENCE WITH NON-HUMANS. Our third generalization, which is ultimately related to the first two (the unmarked limited-control preference and the marked in-control preference with humans), describes the circumstances under which the *se*-marked form of $\pm se$ AC-verbs is preferred even with non-human nominative DPs. As just illustrated in 4c and 5c, the choice between the variants with and without *se* within $\pm se$ AC-verbs is completely free with a non-human DP across the subclasses of verbs just distinguished. However, construing inanimate entities as endowed with some agency is very common across languages (Cruse 1973, Delancey 1985, Fauconnier 2012, Piñón 2001, Koontz-Garboden 2007 a.m.o). If we explicitly ask French speakers to choose the verbal form that makes the non-human more agentive, we expect them to select the variant with *se*. Thus for instance, if asked to choose which of the two forms 6a vs. 6b presents the rose as more responsible for its change, we expect French speakers to choose 6b. This third generalization is what we call the MARKED RESPONSIBILITY PREFERENCE.

(6) *Marked responsibility preference with $\pm se$ anticausatives*

- | | | |
|----|---|--|
| a. | La rose Ø a flétri.
the rose has faded
‘The rose faded.’ | (less responsibility attributed to the rose) |
| b. | La rose s’est flétrie.
the rose SE is faded
‘The rose faded.’ | (more responsibility attributed to the rose) |

We relate this preference to the fact that only the string with *se* allows, besides an anticausative parse, for a semantically reflexive parse, where the sole nominative DP is construed both as a theme and a responsible agent, ‘performing’ its own change.

Concluding this introduction, Table 1 summarizes our generalizations, where the two rows of the ‘Human’ column correspond to the unmarked limited-control preference found with limited-control verbs and the marked in-control preference found with in-control verbs. The ‘Non-human’ column corresponds to the marked responsibility

preference, which arises only when the speaker explicitly aims to present the inanimate as agentive. In brackets, we refer to the experiments presented later on to substantiate these generalizations.⁶

<INSERT TABLE 1 ABOUT HERE>

The remainder of this paper is structured into two parts. Section 2 presents our competition-based lexical pragmatic account of the three generalizations. Section 3 critically discusses a previous prominent claim in the literature, according to which the morphological marking of French anticausatives goes along with a general and systematic meaning differences. We contrast this claim with our three novel generalizations, and present our acceptability rating studies which support these generalizations.

2. A LEXICAL PRAGMATIC ACCOUNT. Understanding the three effects experimentally supported in section 3 (and summarized in Table 1) requires taking into account the multifunctionality of the morpheme *se*. More concretely, *se* is used to form different verbal diatheses (or Voices) which cannot be distinguished on the basis of the surface string. For the above biases, the relevant diatheses are anticausative predicates and semantically reflexive predicates.⁷ A surface string of the form [DP *se* verb] is formally ambiguous between different diatheses, given the REFLEXIVE-ANTICAUSATIVE SYNCRETISM: the clitic *se* is used in French (and in many languages beyond) both as a morphological device forming reflexive verbs (reflexivizer) and as a morphological device forming anticausative verbs (anticausative morphology).

In the next subsection, we first flesh out our assumptions about the semantic interpretations going along with these diatheses. In section 2.2, we then present our account for the three biases. This account only hinges on the existence of a different semantics for each diathesis in competition and the assumption that anticausative structures with or without *se* have the same meaning, and not on the specific syntactic properties assumed to underlie them. Any theory sharing these assumptions (see e.g. Reinhart 2002, Reinhart & Siloni 2005 or Schäfer 2008, Alexiadou et al. 2015 for two different proposals) can equally derive the tendencies we are interested in.⁸

2.1. THE SYNCRETISM OF ANTICAUSATIVE MORPHOLOGY. French alternating verbs can be used as lexical-causative verbs, semantically reflexive verbs (necessarily marked with *se*), and anticausative verbs (marked with or without *se*). We remain here largely noncommittal with regard to the syntax of these different uses and only give the semantics of untensed sentences illustrating the relevant diatheses, which is sufficient for our purposes.

The logical form in 7b provides the semantic interpretation of the sentence 7a involving the agentive, lexical causative use of the predicate *plier* ‘bend/fold’ (we ignore the contribution of tense/aspect). We call the thematic role of the subject DP ‘agent’ and assume that it can be assigned both to human and non-human entity-denoting DPs serving as the external argument of eventive predicates (see Cruse 1973, Fauconnier 2012, Folli & Harley 2005, Koontz-Garboden 2009 a.o.; cf. also the theta-role ‘instigator’ in Borer 2005 or Ramchand 2008).

(7) Transitive verb/lexical-causative verb:

- a. Ana a pli   le guidon. ‘Ana folded/bent the handlebars.’
- b. $\leadsto \lambda e. \exists s(\mathbf{agent}(e, \mathbf{ana}) \wedge \mathbf{cause}(e, s) \wedge \mathbf{bent}(s) \wedge \mathbf{theme}(s, \mathbf{the\ handlebars}))$

Next, the simplified meaning for semantically reflexive construals is illustrated in 8b, which gives the semantics of 8a (again ignoring tense and aspect). The agent can also be an inanimate, as in 8c, the semantics of which is given in 8d. The semantically reflexive reading can be enforced by adding an intensifier.⁹

(8) Semantically reflexive causative verbs:

- a. Ana s’est pli  e. ‘Ana bent.’ (animate agent)
- b. $\leadsto \lambda e. \exists s(\mathbf{agent}(e, \mathbf{ana}) \wedge \mathbf{cause}(e, s) \wedge \mathbf{bent}(s) \wedge \mathbf{theme}(s, \mathbf{ana}))$
- c. Le guidon s’est pli  . ‘The handlebars folded.’ (inanimate agent)
- d. $\leadsto \lambda e. \exists s(\mathbf{agent}(e, \mathbf{the\ handlebars}) \wedge \mathbf{cause}(e, s) \wedge \mathbf{folded}(s) \wedge \mathbf{theme}(s, \mathbf{the\ handlebars}))$

Unmarked anticausative verbs lack any agent-related (or more generally, external argument-related) entailments and denote one-place predicates of change such that the sole DP is interpreted as the undergoer of the event. The anticausative sentences in 9a and 9c (with animate and inanimate undergoer DP) have the semantics in 9b/d (ignoring tense/aspect).

(9) Unmarked AC:

- a. #Ana a pli  . ‘Ana bent.’ (animate undergoer)
- b. $\leadsto \lambda e. \exists s(\mathbf{cause}(e, s) \wedge \mathbf{bent}(s) \wedge \mathbf{theme}(s, \mathbf{ana}))$
- c. Le guidon a pli  . ‘The handlebars bent.’ (inanimate undergoer)
- d. $\leadsto \lambda e. \exists s(\mathbf{cause}(e, s) \wedge \mathbf{bent}(s) \wedge \mathbf{theme}(s, \mathbf{the\ handlebars}))$

With Reinhart 2002, Reinhart & Siloni 2005 or Sch  fer 2008, Alexiadou et al. 2015, Sch  fer & Vivanco 2016 among many others, we assume that anticausatives with *se* also

denote one-place predicates of change. The meaning of 10a and 10b is therefore identical to the meaning of 9a/c given in 9b/d (recall that *plier* is a $\pm se$ AC-verbs).¹⁰

(10) Marked AC:

- a. Ana s'est pliée. 'Ana bent.' (animate undergoer)
- b. Le guidon s'est plié. 'The handlebars bent.' (inanimate undergoer)

To conclude, two different diatheses are realized with the very same surface string [DP *se* verb]: semantically reflexive verbs or *se*-marked anticausatives, and the meaning of *se*-anticausatives can, in principle, also be expressed with unmarked anticausatives.

We now present the pragmatic account of the three preferences presented in section 1, starting with the effects that arise in the context of a human DP with limited-control verbs (the unmarked limited-control preference) and in-control verbs (the marked in-control preference).

2.2. REASONING ABOUT THE CHOICE OF THE ANTICAUSATIVE FORM WITH HUMAN DPs. Our analysis for the two effects involving human DPs is diagrammed in Figure 1 and rests on the assumption that when hearing or reading *se* in a clause headed by a $\pm se$ AC-verb with a human argument, the reflexive parse (under which the DP is associated both with the theme and agent roles) will always be among the salient parses as a consequence of the well-known AGENT BIAS (or AGENT PREFERENCE) in comprehension studies: we tend to preferentially interpret thematic role-ambiguous DPs in subject position as agents (Bickel et al. 2015, Sauppe et al. 2023 and references therein), at least when the DP is human.

The basic idea is that when the language user intends to express an anticausative statement with a human subject and a limited-control or an in-control $\pm se$ verb such as in 4–5, they will choose between the forms with or without *se* so as to manage the ambiguity induced by the *se*-morpheme in the most perspicuous way, following Grice's (1975) Manner supermaxim BE PERSPICUOUS. When faced with a forced choice as the case with $\pm se$ anticausative verbs, being clear/perspicuous amounts among others to avoiding forms that trigger misleading inferences and preferring forms that trigger inferences in line with shared assumptions.¹¹

<INSERT FIGURE 1 ABOUT HERE>

THE UNMARKED LIMITED-CONTROL PREFERENCE. With a human DP and a limited-control $\pm se$ AC-verb such as *rougir* 'redden/blush' in 4, a cooperative speaker will typically avoid the ambiguity and therefore choose the form without *se*, which

unambiguously conveys the (non-agentive) anticausative meaning. For if they chose the ambiguous *+se* form instead, the hearer will reason that the speaker did so because they were after the agentive use (which cannot be conveyed by the unchosen *−se* form). This inference is misleading with limited-control verbs, for that goes against prior shared assumptions about events denoted by these verbs, when such events involve a human.

The hearer’s reasoning upon hearing a (degraded) clause with the *se*-form of a $\pm se$ limited-control verb is decomposed in 11.

(11) *Possible utterances:*

Jeanne rougit.

‘Jeanne is reddening.’

Jeanne *se* rougit.

‘Jean *SE* is reddening.’

- a. The speaker chose *se*.
- b. In a clause with *se* and a change-of-state verb, the (single) human DP argument can either be a mere undergoer, or also an agent of the change they undergo.
- c. The form without *se* would have been a univocal way of expressing that the human DP is a mere (non-agentive) undergoer.
- d. The speaker avoided this form without *se*.
- e. The speaker intended for the agentive meaning.
- f. The speaker aims to signal that the undergoer is an agent of their change.

That is, given that both the variants with and without *se* can in principle be used for the AC reading (which is non-agentive), while only *se* can be used for the reflexive reading (which is agentive), then if the speaker chooses *+se*, the hearer will reason that it is because they are after the reflexive, agentive, reading.

That the effect reflects an optimal choice between forms is supported by the fact that this preference exclusively arises with $\pm se$ AC-verbs (for which there is a choice between forms). The class of *+se* AC-verbs includes limited-control verbs (e.g. *s’affaiblir* ‘weaken’, *s’anémier* ‘become anaemic’), and despite the presence of *se*, they are completely fine with a human DP; see 12, in contrast with 4b. That is, with a human DP, the *+se* form of these verbs can be understood as conveying a change typically not controlled by its theme, despite the fact that the subject is potentially presented as an agent in the grammar, given that the presence of *se* makes a semantically reflexive parse possible. This is because the use of *se* is imposed by grammar for the anticausative use of these verbs and cannot therefore be interpreted as a way to convey the inference that the DP’s referent is agentive.

- (12) *limited-control +se anticausatives*
 Chaïm s'est affaibli/ s'est anémié.
 Chaïm SE is weakened SE is gotten.anaemic
 'Chaïm weakened/got anaemic.'

Degen 2013 and Degen & Tanenhaus 2015 discuss a similar case in the domain of determiners, relying on Manner as we do. As they observe, the vague plural indefinite *some* is suboptimal in cases where a less ambiguous cardinal indefinite like *two* can apply and is a salient alternative. For instance, in a situation where there are two boys, two girls, four socks and four soccer balls, and one of the girls is given two socks while one of the boys is given two soccer balls, 13 is not very natural (and note that in this example, *two* is clearly a salient alternative to *some*, since it is used in the previous clause).¹²

- (13) One of the girls got two of the socks and one of the boys got some of the soccer balls. (Degen & Tanenhaus 2015: 675).

They propose that the problem is due to a misleading determiner choice. In a situation where *two* is a salient alternative to *some* and applies to the situation, the speaker should use it. If the speaker uses *some* instead, the hearer reasons on the basis of the Manner maxim that the speaker must not have in mind a set of two elements. This turns out confusing in a situation where the relevant set counts two elements.

The common point with our case is the hearer's reasoning, which goes as follows: if the speaker faces a choice and chooses the more ambiguous form (*some* for Degen and Tanenhaus's 2015 case, or the *se*-marked verb for us), it is because this speaker is after the meaning which is not conveyable by the less ambiguous form.¹³ In both cases, this choice is confusing as it does not align with shared contextual assumptions.

Of course, it is totally fine to use *some* to describe the same situation above when *two* is not a salient alternative. This is because in such a case, the use of *some* is not interpretable as a choice against the use of the less ambiguous expression *two*, just as the use of the *+se*-form with *+se*-ACs is not interpreted as a choice against the use of the *−se*-form, since the latter is not a grammatical alternative with *+se*-ACs. The misleading inference is only triggered when the use of the more ambiguous form is clearly the result of a choice by the speaker between forms conveying the same meaning.

Still another case where a violation of AVOID AMBIGUITY has been claimed to yield contextual infelicities concerns the competition between reflexive and non-reflexive pronominal forms, illustrated in 14. The relevance of a pragmatic account for the inference

of disjoint reference in 14a based on AVOID AMBIGUITY is discussed in Dowty 1980, Reinhart 1983, Levinson 1987, Roelofsen 2010 and Ivan 2020, among others.

(14) *Context: Felix voted for Felix*

- a. #Felix voted for him.
- b. Felix voted for himself.

The main idea is that while reflexive pronouns as in 14b only have the locally bound interpretation, non-reflexive pronouns as in 14a are ambiguous between a bound and a disjoint interpretation. If a speaker prefers the ambiguous form over the unambiguous one, the hearer reasons that the speaker did not want to convey binding. The disjoint interpretation in 14a is on this view conceived as a manner implicature derived on the basis of AVOID AMBIGUITY. To account for this inference, Dowty 1980 adopts the following principle, which is exactly the one we used to explain the previous cases:

- (15) If a language has two (usually simple) types of syntactic structures A and B, such that A is ambiguous between meanings X and Y while B has only meaning X, speakers of the language should reserve structure A for communicating meaning Y (since B would have been available for communicating X unambiguously and would have been chosen if X is what was intended).

(Dowty 1980: 32, apud Ivan 2020: 53)

An interesting question concerns the high degree of grammaticalization of the optimal choices discussed here. As a reviewer rightly notices, the suboptimal forms for limited-control ACs (i.e. the forms with *se*) do not look like cases of deliberate violation of a pragmatic maxim, but rather like grammatically marked utterances (in a context clearly targeting the anticausative reading). For instance, instances of *se rougir* to mean *blush* collected in Appendix 2 are perceived as atypical ways to speak. But such a situation is, in fact, expected under our assumption that the choice between forms is guided by Manner. Indeed, it is well known that manner implicatures are more prone to conventionalization and grammaticalization (Levinson 2000).

Turning back to reflexive vs. non-reflexive pronouns (cf. 14a/b), it has been similarly noted that a problem for a purely pragmatic account of the inference of disjoint reference in 14b is that this inference does not seem cancellable, which is potentially problematic if it is analysed as an implicature. Roelofsen 2010 notes for instance that a sentence like 16b is not a felicitous continuation for 16a:

- (16) a. John thinks that Felix voted for him.
 b. #In fact, it is possible that John thinks that Felix_j voted for himself_j.

Following up on a suggestion by Roelofsen 2010: 136, Ivan 2020: 55 proposes that the difficulty to cancel the disjoint reference implicature may be due to the fact that the pragmatic inference based on Manner has been conventionalized. Once conversational manner implicatures are conventionalized, we indeed do not expect them to be cancellable anymore.¹⁴

THE MARKED IN-CONTROL PREFERENCE. With a human DP and an in-control $\pm se$ verb such as *plier* ‘bend/fold’ in 5, the most perspicuous way to handle the ambiguity amounts on the contrary to preserving it: the speaker intending to express an anticausative statement with an in-control $\pm se$ verb and a human DP will typically choose the ambiguous variant with *se*. If they chose the variant without *se* instead, the hearer would think that the speaker did so in order to explicitly exclude the reflexive reading. This would, in turn, signal that the DP’s referent is devoid of all agency. This ‘no-agent’ inference goes against shared default expectations about the way human undergoers participate in the changes-of-state denoted by in-control verbs: they are often conceived minimally as ‘doers/effectors’ in such changes. A schematic for the hearer’s reasoning upon hearing a (degraded) clause as in 5a with an in-control $\pm se$ verb but without *se* is given in 17.

- (17) *Possible utterances:*
 Jeanne plie en deux.
 ‘Jeanne is bending in two.’
 Jeanne se plie en deux.
 ‘Jean SE is bending in two.’
- The speaker avoided *se*.
 - In a clause without *se* and a change-of-state verb, the (single) human DP argument can only be a mere (non-agentive) undergoer.
 - The form with *se* could have expressed the same meaning, as well as the agentive (reflexive) meaning.
 - The speaker avoided this form with *se*.
 - The speaker avoided the agentive meaning.
 - The speaker aims to signal that the undergoer is not an agent of their change.

The competing *se*-variant is therefore the optimal choice, as it does not close the door to the agentive (reflexive) parse.¹⁵

We want to underline that an anticausative construal does not by itself entail non-agentivity: there is in principle nothing wrong with the use of an anticausative in agentive situations.¹⁶ But here, the choice of the form without *se* triggers a ‘no-agent’ inference via the reasoning decomposed in 17.

Again, the effect arises because the speaker faces a choice with $\pm se$ in-control verbs: there is, in principle, nothing wrong with presenting the human DP’s referent as an undergoer only in the syntax and semantics (for instance using an anticausative sentence), in a situation where this referent clearly is the agent of their change to some degree. In-control $-se$ ACs (like e.g. *changer de position* ‘change in position’, *entrer* ‘enter’), for which the $+se$ -form is not available in the grammar, are here instructive. When the change of location expressed by the anticausative variants of these verbs involves a human undergoer, it is also often assumed by default that this undergoer has some agentive properties. For instance in 18, the cosmonaut Valentina Tereshkova is typically conceived as contributing to the event as an agent to some degree (a cosmonaut who changes their place in space because of a slight change in pressure ‘does’ something to some degree). But she is encoded as a pure undergoer in the grammar, as the PP enforces the anticausative reading of the verb and no reflexive pronoun is present. The reason for why 18 is unproblematic is simple: since the form without *se* is the only grammatical form for the anticausative of $-se$ ACs, choosing this form is not understood as a choice made to signal the lack of agency of the human undergoer. Therefore, 18 does not trigger the kind of misleading inference making sentences like 5a infelicitous.

(18) *in-control -se anticausatives*

Valentina a changé de position/ est montée/ est descendue de deux mètres
 Valentina has changed of position is ascended is descended of two meters
 à cause d’un léger changement de pression dans le vaisseau spatial.
 because of a light change of pressure in the ship spatial
 ‘Valentina changed in position/ascended/descended by two meters due to a light
 change change in pressure in the spacecraft’.

The reason why a problem arises when the speaker chooses the anticausative form without *se* of $\pm se$ in-control verbs as in 5a is precisely because the speaker makes a choice against the form ambiguous with the (agentive) reflexive meaning: this choice is confusing *because* of the possibility, for $\pm se$ verbs, to form the anticausative with *se*, and in the same time leaving the door open for a reflexive parse, under which the DP’s referent is also an agent.

de Swart and Farkas (2010) discuss a similar case in the domain of determiners.

The common point with our effect is that the most optimal choice is to maximize underspecification (i.e. choose the form maintaining the ambiguity) rather than to use the form univocally expressing the target meaning, and this in order to align with contextual shared assumptions.

Their case concerns the competition between bare plurals (*brooms*) and singular indefinites (*a broom*). As is well-known, plural forms such as *brooms* can in some environments (such as questions) range over sums AND atoms (*Do you have children? Yes, I have a daughter*). By contrast, the singular indefinite has atomic reference only. Now suppose that you want to buy a broom and visit a shop in order to do so. Typically, you want to buy one broom only (atomic reference), but as de Swart and Farkas (2010: 37) rightly observe, the most natural question in such a context is 19b, which contains the ambiguous form, rather than 19a, which unambiguously has atomic reference.

- (19) a. Do you have a broom?
b. Do you have brooms?

As they argue convincingly, the ambiguous, plural form is preferred over the singular one not because the speaker is interested in buying more than one broom, but because by maintaining the referential ambiguity with sums, they align with shared default assumptions, since stores normally sell more than one item of a specific type.

Similarly, the speaker who chooses the *+se*-form of $\pm se$ in-control ACs does so not because they are after the reflexive reading, but because by maintaining the ambiguity with the reflexive reading, they align with shared assumptions about in-control events involving human undergoers.

The two effects we just explored showed us that while the speaker choosing an anticausative construal is in principle neutral about whether the human argument is agentive in the event (after all, ANTI-causative construals are typically used to de-emphasize the entity causing the change), $\pm se$ verbs force the speaker to take a stance on the matter. Choosing the variant with *se* amounts to presenting the human argument as potentially endowed with agency (since the form with *se* is ambiguous); choosing the variant without *se* amounts to presenting this argument as lacking agency (since the form without *se* is unambiguous). With $\pm se$ verbs, neutrality about agency is thus precluded by grammar; the speaker automatically conveys a stance via the (forced) choice. With *-se* or *+se* limited-control or in-control verbs, on the other hand, the presence or absence of *se* does not convey any belief on the part of the speaker about the degree of agency of the human DP, since the morphology is grammatically fixed and therefore does not reflect a choice by the speaker.

2.3. MANAGE (RATHER THAN AVOID) AMBIGUITY AS A SUBMAXIM OF MANNER.

AVOID AMBIGUITY. We proposed that the Gricean submaxim of manner AVOID AMBIGUITY is the pragmatic principle behind the unmarked limited-control preference. But the view that ambiguity avoidance plays a significant role in natural language has been called into question (Wasow 2015 and references therein). Recent research makes clear that ambiguity has many *raisons d'être*: it is a feature of efficient communication systems, allowing a smaller lexicon and better signal compression, among other advantages (Piantadosi et al. 2012, Brochhagen 2018, 2020, Achimova et al. 2022). Furthermore, experimental studies indicate that ambiguity is not such an obstacle to communication for speakers to systematically avoid it (Ferreira et al. 2005). The current consensus seems to be that ambiguity is not avoided ‘across the board’ but in specific situations. In the approach developed in Brochhagen 2018, 2020, for instance, the speaker avoids ambiguity WHEN it decreases the risk of misunderstanding, but not when alternative unintended meanings are nonsensical or when discrimination of the different interpretations does not matter for communication (see also Wasow 2015: 39). The fact that the unmarked limited control preference only arises with a HUMAN argument is in fact a nice illustration of this: speakers avoid the ambiguity triggered by *se* in the context of a human DP because it is perceived as potentially confusing (since whether a human undergoer is also an agent makes a big difference), but are indifferent to the risk of ambiguity with a non-human DP (for the reading where the inanimate undergoer is also an effector typically remains unnoticed because there is no preference to view inanimate subject DPs as agents). Patterns of coexpression in natural language (i.e. expression of different meanings by a single form) seem to obey the same ‘care only when it matters’ logic. For instance, colexification (the coexpression of two lexical meanings in the same word) is more likely to happen when the two meanings are conceptually related, but also when the risk of confusability is not too high (Brochhagen & Boleda 2022).

While it is thus clear that ambiguities are pervasive in natural languages and not systematically avoided by language users, argument structure – who did what to whom – has nonetheless been considered to be an area where ambiguity is characteristically avoided, as it is so central to what is communicated: ‘although it is possible to construct sentences that are ambiguous with respect to argument structure, such examples are the exception rather than the rule’ (Wasow 2015: 36).

The prevalence of voice syncretisms across languages seems to contradict this. For example, Bahrt (2021) finds that 91 (i.e. 41%) of the 222 languages in his sample instantiate voice syncretism with full resemblance in voice marking. Still, the results of

Bahrt’s typological survey prove Wasow 2015 and Brochhagen & Boleda 2022 right in the sense that well-attested voice syncretisms, like for instance the reflexive-reciprocal syncretism, often seem to generate innocuous ambiguities and to coexpress conceptually close meanings. By contrast, the applicative-anticausative syncretism for example, probably disruptive in terms of communication, is never attested in Bahrt’s (2021) language sample, what Bahrt (2021: 120, 237) relates to the conceptual distance between these two voices (since applicatives increase, while anticausatives decrease, the number of participants).

That being said, even if it is certainly relevant, while assessing the relevance of the maxim AVOID AMBIGUITY, to recall that ambiguities are part of the DNA of natural languages, one should bear in mind that Grice sees this principle as a CONVERSATIONAL maxim. This maxim is neither contradicted by the pervasiveness of ambiguous expressions in natural languages (see also Ehrich 2023: 31), nor violated by the use of ambiguous expressions for which there is no unambiguous alternative. Recall for instance that in French, limited-control *+se* ACs raise no problem with a human subject, since the *–se*-form is not available for these ACs (recall example 12). Another example would be a language with causative-applicative syncretism (another well-attested pattern in Bahrt’s sample) such that any causative construal in this language is formally identical with an applicative construal. A speaker of this language producing a causative utterance does not violate AVOID AMBIGUITY, since there is no way for them to convey the target sense with an unambiguous form of the same complexity.

MAINTAIN AMBIGUITY. The marked in-control preference as well as the preference for the plural form in some contexts such as 19 reviewed in section 2.2. illustrate a very different type of case, namely one where preserving an ambiguity serves the communicative purposes better than avoiding it. In that sense, something like MANAGE AMBIGUITY seems more appropriate as a submaxim of manner centered on how ambiguities should be dealt with in cooperative communication. Figure 2 schematises the two optimal strategies of ambiguity management discussed in section 2.2 using the examples of $\pm se$ ACs and determiners.

<INSERT FIGURE 2 ABOUT HERE>

While we are not aware of analyses resorting to a preference for ambiguity to account for infelicitous utterances other than those discussed in section 2.2.2, linguists discuss more and more cases where speakers engage in a deliberate strategy of maintaining ambiguity (Poesio 1996, 2020), for a variety of reasons. Speakers may intend to leave the hearer

uncertain as to the intended interpretation as a rhetorical device (Achimova et al. 2024, see Rathert 2015 for the intentional use of ambiguities in legal texts). Speakers may also intend to communicate more than one meaning (see Grice 1975: 54–55, Lewiński 2021), or choose ambiguous utterances in order to learn about the listener’s belief by observing how they resolve the ambiguity (Achimova et al. 2022, 2025). Another motivation to prefer ambiguity pointed to by Achimova and coauthors and similar to what we have observed in our case studies is to avoid conflict of beliefs in the common ground.

It seems also plausible to us that coexpression in natural languages, required to some extent for a simpler morphosyntax and lexicon (Piantadosi et al. 2012), takes more often place in areas where ambiguity preservation is communicatively efficient. Turning back to the morphological classes of AC in French, the coexpression of anticausativity and reflexive semantics does not always seem to be completely random; in some areas, it takes place when it is particularly convenient for communication to maintain the ambiguity. A first example concerns in-control ACs (like e.g. *plier* ‘bend’). In French, it is easier to find in-control ACs among the *+se* and *±se* classes; that is, *−se* in-control ACs are less frequent (as our lists in 35 below also show). This may be a reflection of a tendency in the French lexicon to favor the ambiguous form for the AC in a situation where not closing the door to the reflexive meaning is the optimal solution, even when it is not a target sense. Beyond French, in-control ACs are often marked with a reflexive across languages (see Geniušienė 1987: 315, who calls these verbs autocausatives; recall footnote 4). A second interesting case concerns psychological anticausatives. In their typological study of psychological verbs constructed with a reflexive marker (e.g. *s’inquiéter* ‘to worry’ in French), Stephenson and colleagues (2024) note that psych-verbs marked with a reflexive, particularly frequent in Romance languages, preferentially express evaluative, social emotions, involving a high degree of conscious appraisal (and such that the Experiencer also plays an active role in their change-of-state). Stephenson and colleagues speculate that this may be due to the influence of reflexive semantics (p. 176). Elaborating on this suggestion, we propose that again, ambiguity preservation via the reflexive marking is the most efficient solution here, since it enables the speaker to present the DP’s referent as an Experiencer, while also leaving open the door to a reflexive interpretation, where the Experiencer is also an agent, as typically assumed in the case of evaluative and social emotions.

Summarizing this section, we proposed that there are two optimal ways to manage ambiguities, namely avoiding or maintaining them, depending on whether the non-target sense clashes or on the contrary is in line with beliefs in the common ground. The unmarked limited-control preference and the marked in-control preference with human DPs both reflect a search for the most optimal way to handle the ambiguity induced by

one of the available forms to express an anticausative with a $\pm se$ verb, namely the $+se$ variant. When *se*-marking is optional and therefore the speaker faces a choice between forms, they will prefer the form reinforcing prior shared assumptions on the events expressed by the verb used in the anticausative statement. With in-control ACs, this is the form with *se* (the ambiguity is preserved to avoid signalling a lack of any agentive properties) while with limited-control ACs, this is the form without *se* (the ambiguity is avoided to avoid attributing agency). Alternative choices are SUBOPTIMAL because they invite the hearer to conclude either that the undergoer of the in-control event is devoid of any agentive property, or that the theme of the limited-control event has such properties. In both cases, this inference is confusing because it is in conflict with prior beliefs in the common ground.

2.4. CHOOSING BETWEEN ANTICAUSATIVES WITH AND WITHOUT *SE* WITH NON-HUMANS. The effects just discussed do not arise with non-human DPs, because the lexical bias of limited-control or in-control verbs is inert with non-humans (the notion of control is typically irrelevant for inanimate agents). Similarly, the reflexive parse does not enter the set of salient parses by default, because there is no agent preference for non-humans. Non-humans can be construed as agents in language, but there is no A PRIORI preference to do so. Thus the anticausative parse remains the most salient and obvious parse for the strings marked with or without *se*. As a result, there is no difference in interpretation between a marked or unmarked form, and the speaker's choice ends up completely uninformative.

That being said, non-human and more specifically inanimate entities are often associated with an agentive thematic role (Cruse 1973, Delancey 1990, Folli & Harley 2005, Koontz-Garboden 2009 a.o.). For instance, across languages, we find agent-introducing 'control' morphologies that are compatible with DPs referring to inanimates, not necessarily with the effect of personification of the inanimate (Fauconnier 2012; see e.g. Jacobs 2011 on control morphology in Salish). Similarly, agentive verbs can be combined with non-human subjects across languages. For example, in English, as in French, unergative verbs like *whistle* can take an inanimate subject (Folli & Harley 2005). Likewise, non-alternating manner verbs of contact like *frapper* 'hit' or *toucher* 'touch' select for an agent (or instrumental) subject (see Fillmore 1970, Cruse 1973 on English), at least when they are used in their eventive (non-stative) meaning.¹⁷ Such verbs, too, can have non-human subject DPs, as illustrated in 20 in French and 21 in English.

- (20) La pierre a frappé la fenêtre.
 the stone has hit the window
 ‘The stone hit the window.’

- (21) a. A rock hit the tree. (Fillmore 1970: 127)
 b. What the bullet did was smash John’s collar-bone. (Cruse 1973: 16)
 c. What did the roses do? They bloomed. (Piñón 2001: 358)

Alternating verbs combined with a non-human external argument can obviously also be used agentively in reflexive construals. This means that for $\pm se$ alternating verbs, it is in principle possible to associate a non-human DP with the role of agent in the syntax/semantics when the verbal form is *se*-marked. It is not, however, when the verbal form is unmarked, since reflexive semantics must be expressed with the reflexive morpho-syntax in French (Reinhart & Siloni 2004, Kayne 1975).

Therefore, if a French speaker aims to endow a non-human entity with agentive properties with a $\pm se$ AC-verb (such as the flower in 22), we expect them to choose the marked variant, because the *se*-marked variant is the only form able to yield a semantically reflexive parse of the clause. Under the latter, the referent of the sole DP is not only assigned the internal theta role of an undergoer, but also the external argument theta role of an agent of the lexical-causative variant of the verb. As a result, it is grammatically encoded as the agent (or effector) of an event.

- (22) La fleur {a flétri / s’est flétrie}.
 the flower has wilted SE is wilted
 ‘The flower wilted.’

This observation (supported by the results of experiment 2 reported in section 3.4) is easy to explain. Only the marked string is compatible with an agentive derivation, different from the anticausative one. Language users effectively endorse a reflexive parse if they are asked to endow the non-human entity with responsibility/agency, considering that the *se*-marked variant is the most effective way to do so, as the agentive semantics is never available for the unmarked form.

Our three lexical pragmatic generalizations about the use of $\pm se$ ACs are substantiated by three experiments presented in the next section. In the same section, we critically discuss a prominent alternative view on French anticausatives, according to which so-called ‘external causation’ leads to ACs marked with *se* and so-called ‘internal causation’ to ACs without *se*. We call this view the CAUSATION CLAIM.

3. THREE LEXICAL PRAGMATIC GENERALIZATIONS ABOUT THE USE OF $\pm se$ ANTICAUSATIVES. Labelle 1992, Labelle & Doron 2010 and Doron & Labelle 2011 suggest that meaning differences distinguish ACs marked with *se* and ACs marked without *se*. According to what we call THE CAUSATION CLAIM, ACs marked with *se* denote an ‘externally caused event’, where some entity different from the sole argument DP is assumed to be the causal force responsible for the coming about of the event. ACs formed without *se* express ‘internally caused events’, such that the sole DP itself is understood as being responsible for the coming about of the event, the latter being conceived as internally driven, that is, ‘as unfolding naturally without obvious external control’ (Labelle 1992: 401). To derive these alleged differences in meaning, fundamentally different syntactic structures have been proposed for ACs with and without *se*. Labelle 1992 argues that ACs marked with *se* are unaccusative, while ACs without *se* are unergative, whereas Doron & Labelle 2011 and Labelle & Doron 2010 propose that both forms are unaccusative but differ substantially in their event decomposition and the position where the lexical root is merged in the structure. While we do not go into the details of these proposals, we point out a crucial point of such syntactic analyses. Since the presence vs. absence of *se* is correlated with different syntactic structures, and since the alleged meaning differences are assumed to be grounded in these different syntactic structures, these proposals wrongly predict these meaning differences not only to hold between the two variants of $\pm se$ AC-verbs, but also globally, between $-se$ AC-verbs and $+se$ AC-verbs. As mentioned above and as we return to below, this is not supported by the empirical picture since the effects of our three generalizations only occur with $\pm se$ AC-verbs.¹⁸

3.1. THE CAUSATION CLAIM. The distinction between EXTERNAL CAUSATION and INTERNAL CAUSATION was originally proposed in Levin & Rappaport Hovav 1995: chapter 3, building on Smith 1970, in order to answer the question of when an intransitive verb has a transitive, lexical-causative counterpart. The idea is that externally caused change-of-state verbs such as English *break* and *open* imply some external cause which brings about the breaking and opening event. The external cause can be, for example, an agent or a natural force (Levin & Rappaport Hovav 1995: 108). While these verbs are assumed to be basically transitive, they allow an intransitive (AC) construal because their external cause argument can be LEXICALLY BOUND at the level of lexical semantic representation and, consequently, is not projected to argument structure and syntax. Internally caused change-of-state verbs such as English *rust*, *decay* and *wilt*, on the other hand, were taken to be inherently intransitive predicates, characterized as describing events where something inherent to the sole argument of the verb has brought about

the eventuality (Levin & Rappaport Hovav 1995: 91). The single test offered for internal vs. external causation is the (non-)existence of a causative counterpart, illustrated in 23-24.¹⁹

- (23) a. The door opened.
 b. John opened the door. (externally caused)
- (24) a. The flower blossomed.
 b. *The gardener/*The sun blossomed the flower. (internally caused)

A number of authors have suggested that when a French anticausative verb is attested in both constructions ($\pm se$ AC-verbs), the change-of-state is presented as EXTERNALLY CAUSED when expressed with *se* and as INTERNALLY CAUSED when expressed without *se* (Bernard 1971, Rothemberg 1974, Burston 1979, Labelle 1992, Labelle & Doron 2010, Doron & Labelle 2011). The idea is that the sole DP is identified as ‘the’ cause of the change (the change is ‘internally driven’), and is consequently presented as responsible for the coming about of the event only if the verb appears without *se*.

This reasoning should explain the alleged contrast between 25a and 25b (examples and judgments from Labelle 1992): A handkerchief cannot be held responsible for its becoming red and, thus, this change cannot be internally driven. The verb must therefore be marked to indicate external causation. By contrast, a human who is blushing is necessarily physiologically co-responsible for their change-of-state, which is conceived as internally driven, and thus the verb must remain without *se*.

- (25) a. Il vit le mouchoir #(*se*) rougir. (externally caused)
 he saw the handkerchief SE redden
 ‘He saw the handkerchief getting red.’
- b. Jeanne (#*se*) rougit. (internally caused)
 Jean SE reddened
 ‘Jeanne blushed/reddened.’

While we agree that the overall optionality that characterizes $\pm se$ AC-verbs like *rougir* is suspended in examples like 25b with a human subject, it is actually not in examples with a non-human subject like 25a. As discussed in the previous section and experimentally supported below, the correct empirical (and, in turn, theoretical) divide is thus between human and non-human undergoers of the change-of-state event, NOT by the distinction between internal and external causation. Furthermore, the effect of a human

argument is not the same across the whole set of $\pm se$ AC-verbs: among $\pm se$ AC-verbs, in-control and limited-control verbs need to be distinguished.

3.2. LIMITED-CONTROL VERBS.

VERB CLASS AND HUMAN UNDERGOER, NOT CAUSATION. According to the causation claim, all $\pm se$ AC-verbs should behave the same and enforce the presence of *se* if the event is characterized as externally caused, while disallowing *se* if the event is internally caused.

Apart from the conceptual problem raised by the distinction between internal/external causation, a further problem for this view is that, under closer scrutiny, only limited-control $\pm se$ AC-verbs such as *rougir* in 25 become problematic with *se*, and this only if their sole argument is human.

More French examples of limited-control verbs include the verbs in 26, all of which denote events which, under their most salient readings, describe changes which are typically not controlled by a human undergoer. In this class, we only put verbs compatible with human subjects, which can in principle exert control on some of the changes they endure.²⁰ Limited-control verbs occur in all three morphological classes of French anticausatives, as the examples in 26a–c show.²¹

(26) Some Limited Control anticausative verbs in French:

- a. $\pm se$ ACs : *(se) brunir* ‘brown’, *(se) foncer* ‘darken’, *(se) noircir* ‘blacken’, *(se) pâlir* ‘turn pale’, *(se) rajeunir* ‘become young’, *(se) rougir* ‘redden, blush’.
- b. $+se$ ACs: *s’affaiblir* ‘weaken’, *s’amaigrir* ‘get thinner’, *s’amoindrir* ‘weaken’, *se fortifier* ‘get stronger’, *s’anémier* ‘become anaemic’, *s’arrondir* ‘put on weight’
- c. $-se$ ACs: *grossir* ‘become bigger’, *maigrir* ‘get thinner’, *grandir* ‘grow’, *vieillir* ‘grow older’

Example 27a is actually fine both with and without *se* (against Labelle’s judgment reported in 25a, but 27b is indeed degraded for us with *se*, in line with Labelle’s (1992) judgment. We called this the unmarked limited control preference (with human DPs).

- (27) a. Le fleuve (se) rougit.
the river SE reddens
‘The river is reddening.’
- b. Jeanne (#se) rougit.
Jean SE reddens
‘Jeanne is blushing/reddening.’

The examples in 28, which all have human subjects with $\pm se$ limited-control verbs, show that the unmarked limited control preference holds irrespective of the type of causation involved. The examples with *se* in 28a–b are predicted to be odd also under the causation claim because they denote spontaneous events (internally caused). But the examples in 28c–d (again with *se*) are equally bad, even though the adjuncts in these examples make it clear that the change expressed by their AC verb is externally caused. 28a–d are all fully acceptable without *se*.

- (28) a. #Djamal s'est beaucoup rajeuni ces derniers temps. (internal cause)
 Djamal SE is a.lot rejuvenated these last times
 Intended: 'Djamal rejuvenated a lot lately.'
- b. #Soumia s'est beaucoup pâlie ces derniers temps. (internal cause)
 Soumia SE is a.lot got-paler these last times
 Intended: 'Soumia became much paler lately.'
- c. #Ada s'est rajeunie grâce à cette nouvelle relation. (external cause)
 Ada SE is rejuvenated thanks to this new relationship
 Intended: 'Ada rejuvenated thanks to this new relationship.'
- d. #Les gens se rougissent sous l'effet de ces lunettes. (external cause)
 the people SE turn.red under the effect of these glasses
 Intended: 'People turn red under the effect of these glasses.' (Zribi-Hertz 1987: 45)

A common point to all the examples in 28 is that the context easily accommodates the default inference triggered by limited-control verbs that the human enduring the change does not control this change. But verbs that, by default, are interpreted as limited-control predicates can also be used in contexts that make explicitly clear that the human DP is associated not only with the theme but also the agent role in the syntax/semantics (the change is then often different from the one described by the default use of the verb; for instance, the AC *rajeunir* by default means *take years off/rejuvenate*, but can also mean *make oneself look younger*). In such semantically reflexive contexts, we predict the opposite pattern than in 28: the marked form of the verb should be preferred to the unmarked one, because reflexive semantics must be expressed with the reflexive marker *se* in French (Kayne 1975, Reinhart & Siloni 2004).²² This is indeed the case; for instance, the reflexive has to appear in the example 29, because it is clear that the adults consciously make themselves look younger. The same example without *se* would be very strange, because the purpose clause requires the subject's referent to be an agent, but the limited-control AC indicates that it is not one (cf. English #*He rejuvenated in order to*

speak with his students).

- (29) Certains adultes vont tenter de #(se) rajeunir pour rentrer en contact avec
 ‘certain adults will try to SE get.younger to enter in contact with
 vos progénitures.
 your children’
 ‘Some adults will try to make themselves look younger in order to enter into contact
 with your children.’ (FrTenten20, horizonnm.fr)

The next set of examples shows that NON-HUMAN DPs are generally acceptable with both morphological variants irrespective of the distinction between internal and external causation. The examples in 30a-b mention the existence of an external cause in a prepositional phrase, and the examples in 30c-d express changes conceived as spontaneous. All these examples, which were taken from corpora and double-checked with additional speakers, are fully acceptable irrespective of whether the AC verb appears marked or unmarked.

- (30) a. Le métal s’est rougi sous l’effet de la chaleur.(+se, external cause)
 the metal SE is reddened under the effect of the warmth
 ‘The metal reddened under the effect of the warmth.’ (Zribi-Hertz 1987: 45)
 b. La pierre avait rougi sous l’effet du feu. (-se, external cause)
 the stone has reddened under the effect of.the fire
 ‘The stone reddened under the effect of the fire.’ (FrTenTen20, chaslerie.fr)
 c. L’air se noircit (...) et la tempête arrive. (+se, internal cause)
 the air SE blackens and the storm arrives
 ‘The air blackens and the storm arrives.’
 (FrTenTen20, www.academie-francaise.fr)
 d. Le papier thermique (...) a tendance à noircir (-se, internal cause)
 the paper thermal has tendency to blacken
 spontanément.
 spontaneously
 ‘Thermal paper tends to get black spontaneously.’ (FrTenTen20, docplayer.fr)

In sum, the data confirms the empirical picture given in section 1: in a default context (i.e. not a semantically reflexive context as in 29), the variant without *se* of limited-control $\pm se$ verbs is very much preferred if the sole argument is human, but both variants can be

used if the sole argument is non-human. The distinction between internal and external causation does not interfere in the distribution of the morphological marking in $\pm se$ ACs.²³ Furthermore, the unmarked limited-control preference for humans does not arise with limited-control $+se$ AC verbs, for which there is no choice between forms either (recall example 12 and the discussion around it).

EXPERIMENT 1A. To test whether the unmarked limited-control preference is robust, we conducted an online acceptability study with native speakers of French (N = 154) (Full details of the experiment can be found in the Appendix 1 and online materials). Participants were asked to read example sentences built with one of the five $\pm se$ limited-control verbs listed in 31 and to rate them for acceptability on a 7-point Likert scale (an additional verb, namely *foncer* ‘darken’, was used with non-human subjects only, as it does not combine naturally with human subjects in French). Distractors were mixed with the test items. An example of the test items is given in Figure 3. The example is translated into English in 33b.

- (31) $\pm se$ limited-control verbs used in Experiment 1a
brunir ‘brown’, *noircir* ‘blacken’, *pâlir* ‘get pale’, *rajeunir* ‘get young(er),
 rejuvenate’, *rougir* ‘redden, blush’

<INSERT FIGURE 3 ABOUT HERE>

The 2x2x3 design manipulated the following factors:

- (32) a. SE: whether the verb of the sentence appeared with *se*-marking or without.
 b. ANIMACY: whether the sole argument was human or not.
 c. CONTEXT: whether the sentence was presented without context (we call this ‘neutral context’), in an inchoative context, or in a semantically reflexive context.

Examples of the three contexts are given in 33 for the verb (*se*) *pâlir* ‘fade, go pale, make oneself pale’ in the context of a human argument and an unmarked version of the verb (example 33c is predicted to be bad due to the absence of *se*, recall 29). Each trial with a human argument contained a proper name in subject position.

- (33) a. NEUTRAL CONTEXT
 Rachida a pâli.
 Rachida has gone.pale
 ‘Rachida went pale.’
- b. INCHOATIVE CONTEXT
 Djamila a pâli à l’annonce de l’infidélité de son amoureux.
 Djamila has gone.pale at the.announcement of the affair of her lover.
 ‘Djamila went pale when she heard about her lover’s affair.’
- c. REFLEXIVE CONTEXT
 Khadija a pâli pour les besoins de son personnage de théâtre.
 Khadija has gone.pale for the needs of her role of theater
 ‘Khadija went pale for her theater role.’

Inchoative contexts were set up with a prepositional causal adjunct specifying an external cause of the change. This context enforces an inchoative/anti-reflexive parse of the clause (an alternative reflexive parse is very implausible across target items; for instance, in 33b, it is very implausible to come up with a scenario where Djamila made herself pale when she heard about her lover’s affair). Reflexive contexts were set up with the help of an adjunct reason clause or a purpose adjunct PP as in 33c, which indicates that the human subject of the main clause is ascribed control over the event (we return to inanimate subjects in reflexive contexts in section 2.4).

With the reason clause or purpose-PP, we enforce a construal where the human sole DP is understood as an external argument. Since no second DP is available that could be interpreted as the internal undergoer argument, the most plausible parse is one where the sole DP is both the external and the internal argument, thus a reflexive interpretation. Given that in French, a reflexive interpretation is obligatorily marked with the clitic *se*, we predict the variant with *se* to be rated high and the variant without *se* to be rated low in a reflexive context.

Based on the discussion above, our predictions were as follows:

- (34) a. With human arguments, the variant without *se* will be rated higher than the variant with *se* in the neutral and inchoative contexts. Non-human arguments will not show this preference. This is our unmarked limited-control preference for humans.
- b. With human arguments, the variant with *se* will always be rated higher than the variant without *se* in the reflexive context. With non-human arguments, we do not expect such a difference in the ratings of forms with and without *se* (we

come back to this point in section 3.4).

<INSERT TABLE 2 ABOUT HERE>

The results are summarized in Table 2, which gives raw means and standard errors for each condition, and in Figure 4, where each dot indicates a single trial (one sentence rated by one participant) and error bars give 95% confidence intervals. The confidence intervals can be understood as follows: if we ran the same experiment many times, we expect the mean rating to fall somewhere between these error bars 95% of the time; this ‘spread’ of values gives a better indication of uncertainty than a single mean value (the sample mean can still be seen in the black dot halfway between the two ends of the error bars). Informally, when the error bars of two conditions do not overlap, this is evidence that the two conditions differ. So for example, in the Inchoative and Neutral panes, looking at human subjects, there is evidence that participants rate examples without *se* substantially higher than sentences with *se*. In these two panels, most ratings are high for the no-*se* conditions, but more varied and negative overall for the yes-*se* condition. By contrast, turning to non-human subjects in the same Inchoative and Neutral panes, there is no visible difference in the ratings for sentences with and without *se* (the ratings and error bars for the two conditions overlap). In the Reflexive pane with human subjects, participants rated the forms with *se* higher than the forms without *se*, which were negative overall. This difference is again not observed with non-human subjects in the same reflexive pane. The individual dots reflect the overall variation in our sample.

<INSERT FIGURE 4 ABOUT HERE>

These findings were evaluated using an ordinal Bayesian analysis (see Appendix 1 for full model output and the online repository for additional confirmatory analyses, including ROPE, emmeans and Bayes Factor, all consistent with our findings).²⁴ Our prediction was that we would see lower ratings when human arguments have *se* in the neutral and inchoative contexts. The relevant effects whose estimate is reliably different from zero are given in Table 3.

ANIMACY and SE were sum coded. CONTEXT was treatment coded with Neutral as the baseline level (the intercept). The model’s 95% Credible Interval for the interaction of Human and *se* lies in the range $[-3.58, -2.15]$, meaning that a *se*-marked human clause is rated almost 3 likert points less (estimate = -2.87) than a human clause without *se*, before considering Context. This effect is then immediately qualified by additional interactions; we simplify slightly now by focusing on the three-way interaction between Animacy, Se

and Context. The effect just mentioned means that examples with human arguments receive lower ratings when they have *se*, but this assumes the baseline context Neutral. In Inchoative contexts (interaction of Human:Se:Context), this effect is about 1.5 likert points less strong than in the Neutral context, but still substantial overall. The pattern is therefore confirmed for the Inchoative and Neutral contexts, as predicted. In other words, this is the human unmarked limited-control preference. The three-way interaction of Human:Se:Inchoative indicates that the effect is slightly stronger in the Neutral context than the Inchoative one, a pattern that can be seen in Figure 4 as well, and not one we had any prior hypotheses for.²⁵

<INSERT TABLE 3 ABOUT HERE>

The next prediction concerns the behavior of human and non-human arguments in the reflexive context. Here we predicted no difference for non-humans, but a preference for *se* for humans. This is what we found: the interaction of Se and Reflexive was not reliably different from 0, meaning there was no difference for non-humans whether they had *se* or not. However, the effect for human arguments was robust: the three-way interaction of Human:Se:Reflexive had a very high estimate (4.82), reflecting the preference of *se* for reflexive clauses with humans, effectively reversing the patterns discussed above when Neutral was the reference level. Additional inferential statistics, described in the Appendix 1 and OSF repository, show for example that the model would predict high ratings for no-*se* sentences with humans in the non-reflexive contexts.

In sum, we found evidence for the unmarked limited-control preference with limited-control verbs. These $\pm se$ limited-control verbs, such as *rougir* ‘blush’, *rajeunir* ‘rejuvenate’ and *pâlir* ‘get pale’ remain preferably unmarked when used as anticausatives with human subjects. This preference does not hold with a non-human subject. We next carry out the same exercise with the second set of $\pm se$ AC-verbs, namely in-control verbs.

3.3. IN-CONTROL VERBS.

VERB CLASS AND HUMAN UNDERGOER, NOT CAUSATION. The limited-control verbs of the previous section contrast with in-control $\pm se$ verbs, which denote changes a human undergoer typically exerts some control over. More examples of this class of in-control anticausative verbs are given in 35.²⁶ These predicates are typically used as motion or posture verbs when combined with a human subject (called autocausatives in Geniušienė 1987 and Creissels 2006, endo-reflexives in Haspelmath 1987 and quasi-reflexives in Creissels 2024, cf. footnote 4), as well as degree achievements expressing a behavioral

change, such as the last five verbs in 35a taken in their behavior-related use. The relevant use is exemplified in 38b and 39b.

(35) Some In-Control anticausatives in French:

- a. $\pm se$ ACs: *(s')* *allonger* ‘get longer/lie’, *(s')* *approcher de* ‘get close(r) to’, *(s')* *avancer* ‘move forward’, *reculer* ‘step backwards, diminish’, *(se)* *plier* ‘bend’, *(se)* *radoucir* ‘soften’, *(se)* *balancer* ‘swing, rock’, *(s')* *arrêter (de marcher)* ‘stop (walking/working)’, *(se)* *courber* ‘bend, curve’, *(se)* *loger* ‘fit, stay’, *(se)* *nicher* ‘squeeze in, tuck oneself in’, *(se)* *durcir* ‘harden’, *(se)* *raidir* ‘stiffen, harden’, *(se)* *ramollir* ‘melt/soften’, *(se)* *refroidir* ‘get cold(er)’, *(se)* *dégeler* ‘unfreeze’
- b. $+se$ ACs: *s’abaisser* ‘get lower, bend’, *se lever* ‘raise, stand up’, *se déplacer* ‘move’, *se mouvoir* ‘move’, *se rapprocher* ‘get closer’, *se relever* ‘got up, get back on one’s feet’, *se retourner* ‘turn over, around’
- c. $-se$ ACs: *bouger* ‘move’, *remuer* ‘move’, *changer (de place)* ‘change (one’s position)’, *monter* ‘climb, go up’, *plonger* ‘dive into, get immersed’, *entrer* ‘get in’

A natural construal for in-control verbs with a single human DP is the reflexive, agentive use, and since reflexive semantics requires an overt reflexive marker in French, the *se*-form must be selected by a speaker who intends to convey this reading. But we are here interested in the case where in-control verbs enter into an anticausative construal, where the human DP is only encoded as an undergoer in the grammar. This is for instance the use selected for *se plier* ‘SE bend’ in 36c when it is used as an answer to 36a. Note that the exchange 36a–c sounds more felicitous than the exchange 36b–c, which is unsurprising given the choice of the unaccusative verb *tomber* ‘fall’ and the modification of *se plier* by the cause-PP *de douleur* ‘from pain’.

(36) ...Et alors Judy a tiré sur Jim.

‘...And then Judy shot Jim.’

- a. Oh wow, et qu’est-ce qu’il lui est arrivé?
‘Oh wow, and what happened to him?’
- b. Oh wow, et qu’est-ce qu’il a fait?
‘Oh wow, and what did he do?’
- c. Il est tombé à genoux et s’est plié de douleur.
he is fallen to knees and SE has bent from pain
‘He fell to his knees and bent over in pain.’

As already mentioned in section 1, with in-control $\pm se$ ACs and human subjects, it is the unmarked form which is problematic. This is what we called THE MARKED IN-CONTROL PREFERENCE (for humans), illustrated with more examples 37–39 below. In the (a)-examples a non-human subject is fine with or without *se*; in the (b)-examples, a human subject is fine with *se*, and the (c)-examples show the degradedness of human subjects in the absence of *se*. We add a cause-PP across examples to enforce the inchoative reading.

- (37) a. La tôle $\emptyset$ a plié/ s'est pliée en deux (sous le poids).
 the metal sheet has folded SE is folded in half under the weight
 'The metal sheet folded in half under the weight.'
- b. Jeanne s'est pliée en deux (de douleur).
 Jeanne SE is bent in two from pain
 'Jeanne bent over (in pain).'
- c. #Jeanne $\emptyset$ a plié en deux (de douleur).
 Jeanne has bent in two from pain
 Intended: 'Jeanne bent over (in pain).'
- (38) a. Ici le temps $\emptyset$ a radouci/ s'est radouci avec l'arrivée de
 here the weather has gotten-milder SE is gotten-milder with the arrival of
 l'été.
 the summer
 'Here the weather got milder with the start of the summer.'
- b. Xiao s'est radouci sous la pression et a libéré les pratiquantes.
 Xiao SE is gotten-milder under the pressure and freed the churchgoers
 'Xiao mellowed under the pressure and freed the churchgoers.'
- (fr.clearharmony.net)
- c. #Xiao $\emptyset$ a radouci sous la pression.
 Xiao has gotten-milder under the pressure.
 'Xiao mellowed under the pressure.'

- (39) a. Le métal Ø a durci/ s'est durci sous la chaleur.
 the metal Ø has hardened SE is hardened under the heat
 'The metal got hard with the heat.'
- b. Laeticia Hallyday s'est durcie après la mort de Johnny.
 Laeticia Hallyday SE is hardened after the death of Johnny
 'Laeticia Hallyday became harder after Johnny's death.' (leparisien.fr)
- c. #Après la mort de Johnny Hallyday, Laeticia Ø a durci.
 after the death of Johnny Hallyday Laeticia has hardened
 Intended: 'After Johnny Hallyday's death, Laeticia became harder.'

The intuitions reported in 37–39 were also tested in an online acceptability study to which we turn next.

EXPERIMENT 1B. Experiment 1b was carried out with the same participants as Experiment 1a (N = 154) during the same session, though items were counterbalanced across participants (see the Appendix 1 and online materials). The experimental setup was the same, except that we used five in-control verbs and appropriate contexts, within the same 2x2x3 design. Verbs used in Experiment 1b are listed in 40 (distractors were again mixed with the test items).

- (40) $\pm se$ in-control verbs used in Experiment 1b
approcher de 'get close(r) to', *durcir* 'harden', *plier* 'bend', *radoucir* 'get soft(er)',
refroidir 'get cold(er)'

Our predictions were as follows (predictions about non-human arguments are discussed in more detail in section 3.4):

- (41) a. For human arguments, the marked variant will be preferred across all contexts (neutral, inchoative and reflexive contexts). (This is our marked in-control preference for humans.)
- b. Non-human arguments will not show this preference: both forms will be acceptable across contexts.

<INSERT TABLE 4 ABOUT HERE>

Raw means and standard errors are given in Table 4. The results are summarized in Figure 5, where each dot indicates a rating and error bars give 95% confidence intervals. Figure 5 shows that in the Inchoative and Neutral panes, looking at human subjects, there

is evidence that participants rate examples with *se* higher than sentences without *se*. Most ratings are high for the yes-*se* conditions, but more varied and negative overall for the no-*se* condition. By contrast, turning to non-human subjects in the same Inchoative and Neutral panes, there is no obvious difference in the ratings for sentences with and without *se* (both forms receive positive ratings overall, confirming that the verbs tested are $\pm se$ ACs). In the Reflexive pane with human subjects, participants rated again the forms with *se* higher than the forms without *se*. This difference is again not observed with non-human subjects in the same reflexive pane (we come back to this last point in section 3.4).

<INSERT FIGURE 5 ABOUT HERE>

These findings were evaluated using an ordinal Bayesian analysis (see the Appendix 1 for full model output and the online repository for additional confirmatory analyses). Our prediction was that we would see higher ratings when human arguments have *se* in all three contexts. The relevant effects whose estimate is reliably different from zero are given in Table 5, with the full output reproduced in Appendix 1 (and additional confirmatory analyses with ROPE, emmeans and Bayes Factor in the online repository).

The model’s 95% Credible Interval for the interaction of Human and *se* lies in the range [0.93,2.21], meaning that a *se*-marked human clause is rated one and a half likert points higher (estimate = 1.57) than a human clause without *se*, across contexts. This effect is qualified by additional interactions; we simplify by focusing on the three-way interaction between Animacy, Se and Context. The effect just mentioned means that examples with human arguments receive higher ratings when they have *se*; the interaction of Human:Se:Context shows that the same happens in Inchoative, which is not reliably different from Neutral (the 95% Credible Interval covers zero, meaning the low estimate of 0.29 is not particularly strong). In other words, there is no difference between Inchoative and Neutral, unlike in Experiment 1a. The general pattern is therefore confirmed, as predicted. This is the human marked in-control preference.

<INSERT TABLE 5 ABOUT HERE>

The next prediction concerns the behavior of human and non-human arguments in the reflexive context. Here we predicted no difference for non-humans, but a preference for *se* for humans. This is what we found: the interaction of Se and Reflexive was not reliably different from 0, meaning there was no difference for non-humans whether they had *se* or not. Additional inferential statistics and model predictions are available in the OSF repository.

In sum, this section provided evidence for the marked in-control preference with in-control verbs. It also showed that the distinction between in-control and limited-control ACs (anticipated by authors such as Geniušienė 1987, Creissels 2006 or Haspelmath 1987) is crucial, as these verbs give rise to opposite patterns WITH HUMAN SUBJECTS. With non-human subjects, the distinction between these two subclasses is largely irrelevant: both forms are accepted across contexts, which also confirm that the verbs at hand are $\pm se$ ACs.

The marked in-control preference is the opposite of what the causation claim predicts (but remember that proponents of this claim did not distinguish between limited-control and in-control verbs like we do). The causation claim says that across ACs, the presence of *se* generally characterizes the event as being externally caused. But the marked in-control preference confirmed by the results of Experiment 1b shows that that *se* is favored when there is a shared assumption that the sole human argument probably exerts some agency in the event they undergo.

In the next section, we evaluate our third effect against one final set of data involving non-human arguments.

3.4. AGENCY AND NON-HUMAN ARGUMENTS. Recall the marked responsibility preference with non-human DPs: given pairs such as those in 42-43, we expect participants to be more likely to choose the marked variant if explicitly asked to attribute responsibility to the subject, as responsibility is a key property of agents or effectors. The reason for this is that while the unmarked variant of the verb only has an anticausative parse, the variant with *se* allows besides its anticausative parse a semantically reflexive parse. But if, as suggested by the causation claim, the absence of *se* indicates greater responsibility of the subject for the event, we expect the opposite choice.

- (42) a. La rose $\emptyset$ a flétri.
 the rose has faded
 ‘The rose faded.’
 b. La rose s’est flétrie.
 the rose SE is faded
 ‘The rose faded.’
- (43) a. Le métal $\emptyset$ a rouillé.
 the metal has rusted
 ‘The metal rusted.’

- b. Le métal s'est rouillé.
the metal SE is rusted
'The metal rusted.'

EXPERIMENT 2. This prediction was also tested in an online study (see again the Appendix and online materials). $N = 33$ native speakers of French participated in the experiment, none of whom participated in Experiments 1a/1b. They were given 12 minimal pairs like those in 42a–b and 43a–b and asked which of the two sentences assigns more responsibility to the subject (*Quelle forme attribue le plus de responsabilité à la rose/au métal dans le procès?* 'Which form assigns more responsibility to the rose/the metal in the event?'). Judgments were provided on a 7-point scale with the two sentences at the extremes (1 for unmarked, 7 for marked, although the scale was not labelled). Participants were introduced to the responsibility scale through training items, translated under 44. An example of test items is given in Figure 6 ('The sugar caramelized').²⁷

<INSERT FIGURE 6 ABOUT HERE>

- (44) Q: On essaie. Quelle forme attribue le plus de responsabilité à la grand-mère/Yining dans le procès?
'Let's try. Which form assigns more responsibility to the grandma/Yining in the event?'
- a. La grand-mère donne à manger au bébé/ Le bébé mange à côté de la grand-mère.
'The grandma feeds the baby/ The baby is eating next to the grandma.'
- b. Yining et Jinhong réparent le bateau ensemble/ Jinhong et Yining réparent le bateau ensemble.
'Yining and Jinhong are repairing the boat together/ Jinhong and Yining are repairing the boat together.'

Our test items were formed with the verbs listed in 45a–c. While these verbs come from different sub-classes, we did not expect these classes to matter in the responsibility attribution: the responsibility effect should hold across all subtypes of $\pm se$ ACs, as the ambiguity of the reflexively marked form is exactly the same across subclasses.²⁸ All verbs have transitive uses beyond their intransitive uses, and thus can enter reflexivization. Verbs in 45a are examples of internally-caused change-of-state verbs (cf. Wright 2002 and see footnote 19). Those in 45b and 45c are a subset of verbs used in Experiments 1a and

1b. Thus when applied to human arguments, verbs in 45b are limited-control verbs and those in 45c are in-control verbs, but this difference is irrelevant for non-human subjects.

Furthermore, 8 pairs of distractors were mixed with the test items. The task was the same, but this time participants had to choose either between a lexical-causative statement (*Hamida a bougé la chaise* ‘Hamida moved the chair’) and the corresponding –*se* anticausative statement from the verbs in 46a (*La chaise a bougé* ‘The chair moved’); or between a *se*-passive sentence (*La voiture s’est lavée au garage* ‘The car *se*-is washed in the garage’) and a corresponding periphrastic passive sentence from the verbs in 46b (*La voiture a été lavée au garage* ‘The car was washed in the garage’). The question for distractors was the same as for test items (*Which sentence assigns more responsibility to the chair/the car in the event?*).

(45) Verbs used in the test items of Experiment 2:

- a. ‘internally caused’ verbs: (*se*) *caraméliser* ‘caramelize’, (*se*) *fâner* ‘wilt, decay’, (*se*) *flétrir* ‘wilt, decay’, (*se*) *rouiller* ‘rust’.
- b. Verbs from Experiment 1a: (*se*) *brunir* ‘turn brown(er)’, (*se*) *foncer* ‘darken’, (*se*) *rajeunir* ‘get younger’, (*se*) *rougir* ‘redden’.
- c. Verbs from Experiment 1b: (*se*) *baissér* ‘lower’, (*se*) *durcir* ‘harden’, (*se*) *plier* ‘bend’, (*se*) *refroidir* ‘get cold(er)’.

(46) Verbs used in the filler items of Experiment 2:

- a. Alternating verbs: *bouger* ‘move’, *brûler* ‘burn’, *fondre* ‘melt’, *ramollir* ‘soften’.
- b. Non-alternating verbs: *laver* ‘wash’, *nettoyer* ‘clean’, *jeter* ‘throw’, *tuer* ‘kill’.

The results of the experiment are summarized in Figure 7, where it can be seen that participants predominantly used the points at the *se*-marked half of the scale, as opposed to the unmarked one, when choosing the form assigning more responsibility to the subject.

An ordinal Bayesian model confirmed the tendency to pick the marked form as the one assigning more responsibility to the single (non-human) DP; this intercept-only model can be found in the Appendix 1. Results did not differ considerably between verb types, although the preference might be slightly weaker with limited-control verbs; see the OSF repository for post-hoc analyses including model fits.

<INSERT FIGURE 7 ABOUT HERE>

EXPERIMENTS 1A/B WITH NON-HUMAN SUBJECTS. We now come back to Experiments 1a/b, more specifically to the condition with a non-human subject in the reflexive context

(see sections 3.2 and 3.3 for the predictions for the inchoative/neutral contexts with a non-human subject). Recall that in Experiments 1a/b, participants were just asked to rate the acceptability of sentences; they were not asked to choose which sentence attributes the most responsibility to the non-human entity. Sentence 47 is an example of a test item with a non-human subject and limited-control verb (Experiment 1a), and 48 is an example built with an in-control verb (Experiment 1b).²⁹

- (47) Dans cette situation, la carapace de l'insecte (se) noircit pour
 in this situation the carapace of the.insect SE blackened in order to
 échapper aux prédateurs.
 escape the predators
 'In this situation, the insect's carapace turns black in order to escape the predators.'
- (48) Certains chargeurs solaires (se) plient pour mieux s'incorporer dans un
 some chargers solar SE fold.up in order to better SE.incorporate in a
 sac à dos.
 bag to back
 'Some solar chargers fold up in order to better fit into a backpack.'

Differently from what we observed with human subjects, the reason clause does not force an agentive construal for the referent of the matrix subject. Neither 47 nor 48 force the charger or the carapace to be construed as external arguments. The reason clause can be understood as just giving the teleological explanation for why the event described in the matrix clause holds. It does not have to be interpreted as the motive behind the behavior of the subject's referent (which therefore does not have to be interpreted as an agent). This conforms to what has been repeatedly observed for English for examples such as 49 (cf. Williams 1974, Williams 2005, Bhatt & Pancheva 2017).

- (49) Grass is green to promote photosynthesis.
 (Williams 1974, cited in Williams 2005: 293)

Given that the reason clause does not enforce an agentive construal with a non-human subject, it does not trigger a reflexive construal of the matrix clause. We thus did NOT expect a preference for the reflexively marked form in the reflexive context with non-human subjects. Results summarized in Figures 3 and 4 confirmed this prediction.³⁰

3.5. EXPERIMENTAL RESULTS: SUMMARY AND DISCUSSION. In summary, what the experimental results show us is that with non-human subjects, the morphological marking in $\pm se$ verbs remains uninformative and unconstrained (Experiments 1a/b): participants equally like both forms across contexts with non-human subjects. However, if the speaker aims to present the non-human entity as agentive and responsible for the change it endures, they will favour the reflexively marked form over the unmarked form (Experiment 2). As for human arguments, results of Experiment 1a on limited-control verbs show that participants like the form without *se* in the neutral or inchoative contexts, and on average do not like the form with *se* in the same context. The opposite pattern is observed in the reflexive context. Again with human subjects, results of Experiment 1b on in-control verbs show that participants like the form with *se* across contexts, and do not like the form without *se* across contexts.

We can further observe that in both experiments 1a/b, the ratings of the unexpected form with human subjects are not only more negative but also appear to be more varied overall than the ratings for the expected form: while the ratings for the preferred forms follow a roughly normal distribution around a high rating (that is, they are centered around this high rating), the ratings for the dispreferred forms are more uniform (i.e. responses are found throughout the scale).³¹

While evaluating this idea properly is a matter for future work, we take this indicative pattern to support the view that the unmarked limited-control preference is pragmatic in nature: the high level of variation in the ratings for the suboptimal forms suggest that some participants ignore to some extent the information conveyed through the choice between forms. In particular, some participants seem to accept the $+se$ -form with limited-control verbs and a human subject. Importantly, these higher ratings cannot reflect a semantically reflexive parse for the *se*-marked form, for as we mentioned in section 3.2.2, this reading is implausible in the target items. For instance, it is unlikely that participants come to a reflexive reading for 50, as it would require that Djamila agentively controlled her change of skin color as a result of hearing about her lover’s affair.

(50) #Djamila s’est pâlie à l’annonce de l’infidélité de son amoureux.
 Djamila SE is gone.pale at the.announcement of the affair of her lover
 ‘Djamila went pale when she heard about her lover’s affair.’

Another relevant point is that the unexpected form is occasionally attested in corpora, suggesting that this form is suboptimal but not plainly ungrammatical. Examples 51 below are attested examples of native speakers of French where the limited-control $\pm se$ verb very

clearly instantiates the anticausative use while they are used with *se* (see the Appendix 2 for more examples).

- (51) a. Son armée avait les yeux braqués sur lui, tandis que sa garde rapprochée
 his army had the eyes turned on him whereas that his guard close-knit
 l'aida à se relever alors que la plupart du reste de la troupe
 him helped to SE stand up while that the most of the rest of the troop
se pâlit à la vue d'un tel spectacle.
 SE got pale to the view of a such spectacle
 'His army had all the eyes on him, and whereas his bodyguard helped him to
 his feet while most of the rest of the troops turned pale at the view of such a
 spectacle.'
madelle.1fr1.net/t2050p10-event-4-le-tremblement-de-la-guerre
 b. J'ai les yeux dans mon café (...) et quand je relève les yeux, elle **se**
 I have my eyes in my coffee and when I raise the eyes she SE
rougit et se détourne.
 reddens and SE turns away
 'I keep my eyes in my coffee (...) and when I raise my eyes she gets red and
 turns away.' forum.canardpc.com/archive/index.php/t-56652-p-18.html

Similarly, the results of Experiment 1b on in-control verbs suggest that some participants simply ignore the confusing 'no-agent' inference triggered by the choice of the unambiguous *–se* variant of the anticausative: while ratings for the form without *se* of in-control verbs in the neutral or inchoative contexts are overall negative, they also show a lot of variation. Furthermore, this suboptimal variant of in-control $\pm se$ verbs is also occasionally found in corpora. Examples 52 below are attested cases where the in-control $\pm se$ verbs *plier* 'bend' and *approcher de* 'get close(r) to' are used as anticausatives without *se* in the presence of a human subject (see again the Appendix 2 for more examples).

- (52) a. À ce moment, j'ai saisi la première chose que j'ai vue et je
 at that moment I have grasped the first thing that I have seen and I
 l'ai frappé. Il a plié en deux, mais il a lâché un sacre
 him have hit he has bent in two but he has dropped a curse word
 avant de me donner un coup de poing.
 before to me give a punch
 'At that moment, I grasped the first thing I saw and hit him. He bent in two,
 but he dropped a curse word before giving me a punch.'
 (www.lecitoyenrouynlasarre.com)
- b. dès qu'il approche d'un homme, il crie sans relâche: Ne me
 as soon as he gets.closer of a man he shouts without stopping NEG me
 touchez pas!
 touch NEG
 'As soon as he gets closer to a man, he shouts without stopping: don't touch
 me!' (Gaston Paris, 1880, *Légendes du Moyen-Age*)

Interestingly, it seems that in situations similar to those explored experimentally in Joo et al. 2024 where the human DP is fully incapacitated, for instance completely unconscious or in a coma, the version without *se* becomes preferred over the version with *se*; see for instance 53.

- (53) Paul est très rapidement entré dans le coma quand le camion a foncé
 Paul is very quickly entered in the coma when the truck has crashed
 dans la voiture, et ensuite a plié en deux/ #s'est plié en deux sous le
 in the car and then has bent in two SE is bent in two under the
 poids de la tôle du camion.
 weight of the sheet metal of.the truck
 'Paul quickly fell into a coma when the truck crashed into the car, and then folded
 in half under the weight of the truck's metal.'

We can easily account for why maintaining the ambiguity with the agentive reflexive reading is odd: Paul is here not even a performer of his change of position. The unmarked form is therefore more appropriate in this context, as the 'no-agent' inference triggered by this choice fits perfectly with the default assumptions in a context of full incapacity.

4. CONCLUSIONS. In this paper, we looked at the ways lexical-semantic biases of verbal predicates interact with general conversational principles, focusing on the Manner

supermaxim BE PERSPICUOUS. Starting with the assumption that French anticausatives marked with *se* or left unmarked do not differ in meaning, we examine how cooperative language users are guided in their choice between the marked and unmarked forms. We showed that while the choice between the forms is completely uninformative when the (unique) overt DP is non-human, it becomes strategic when the DP is human. The reason why this choice becomes laden with consequences when the DP is human is that when syntax makes several parses available such that one of this parse associates the role of agent to the human DP, then this parse is always among the salient parses (this is the AGENT BIAS). We argued that in such situations, a cooperative speaker will handle the ambiguity of *se* with perspicuity, in line with the Manner supermaxim BE PERSPICUOUS. This amounts primarily to aligning with shared assumptions about events denoted by the VP. In this respect, we argued that two subclasses of verbs are particularly relevant: limited-control verbs express events that tend to be understood as not under the control of a human undergoer, while in-control verbs express events that tend to be understood as under the control of a human undergoer. Aligning with these shared assumptions means preferring the unmarked variant of limited-verbs, to avoid an ambiguity with the semantically reflexive reading which endows the human DP with agency, but preferring the *se*-marked variant of in-control verbs, to maintain the ambiguity with the reflexive reading and as such avoid triggering the inference that the human DP is not agentive at all. These two preferences (the unmarked limited-control and the marked in-control preferences) are only at play with verbs for which there is a choice between form ($\pm se$ verbs), which supports our view that these preferences result from a pragmatic reasoning on the basis of general Gricean principles of conversation. We call such effects lexical pragmatic effects. While our empirical study was based on French, we expect related effects in other languages once they show diathesis syncretisms and optionality in the morphological realization of particular diatheses.

A. APPENDIX 1: EXPERIMENTAL DESIGN. This appendix contains additional details on our experimental setup. Data from both experiments and the analysis script can be found in the OSF repository on https://osf.io/4jqhn/?view_only=aafec40636bd468eaa3c52b4cf7691e4.

A.1. EXPERIMENT 1.

PARTICIPANTS. Participants were recruited on Prolific and paid EUR 1.70 for participation. All participants self-reported as native speakers of French aged 18 or over, and born in a Francophone European country (France, Belgium, Switzerland). Since we

had no hypotheses about variation, no demographic information was collected. A total of $N = 154$ (161 before exclusions) participants took part, divided randomly into four lists for counterbalancing purposes (A: 37, B: 36, C: 39, D: 42). Experiments 1a and 1b ran in the same session.

PROCEDURE. Participants rated the acceptability individual sentences on a 7-point Likert scale. The radio button on one edge was labeled *Pas du tout naturelle* ‘not natural at all’ and the opposite one was labeled *Tout à fait naturelle* ‘completely natural’. Materials were presented visually using PCibex (Drummond 2007, Zehr & Schwarz 2018). Four practice trials preceded the main experiment, in which the order of trials was randomized.

MATERIALS. Experiment 1 was comprised of two verb classes, limited-control in Experiment 1a and in-control in Experiment 1b. Six limited-control verbs were used in Experiment 1a and five in-control verbs were used in Experiment 1b. In each of these two sub-experiments, items were constructed by crossing three conditions: ANIMACY, SE and CONTEXT.

ANIMACY indicated whether the subject was human or non-human:

- (54) a. Adèle a rougi sous l’effet des moqueries et de l’humiliation.
 Adèle has reddened under the effect of.the teasings and of the shame
 ‘Adèle got red under the effect of the teasing remarks and the shaming.’
 b. L’eau a rougi à cause du sang sur ses mains.
 the water has reddened because of.the blood on his hands
 ‘The water got red because of the blood on her hands.’

SE indicated whether *se*-marking appeared or not:

- (55) a. Adèle a rougi sous l’effet des moqueries et de l’humiliation.
 Adèle has reddened under the effect of.the teasings and of the shame
 ‘Adèle got red under the effect of the teasing remarks and the shaming.’
 b. Adèle s’est rougie sous l’effet des moqueries et de l’humiliation.
 Adèle SE is reddened under the effect of.the teasings and of the shame
 ‘Adèle got red/made herself red under the effect of the teasing remarks and the shaming.’

CONTEXT coded whether the verb was placed in neutral, anticausative or reflexive context, repeated here from 33 in the main text:

- (56) a. NEUTRAL
 Rachida a pâli.
 Rachida has gone.pale
 ‘Rachida went pale.’
- b. INCHOATIVE
 Djamila a pâli à l’annonce de l’infidélité de son amoureux.
 Djamila has gone.pale at the.announcement of the affair of her lover
 ‘Djamila went pale when she heard about her lover’s affair.’
- c. REFLEXIVE
 Khadija a pâli pour les besoins de son personnage de théâtre.
 Khadija has gone.pale for the needs of her role of theater
 ‘Khadija went pale for her theater role.’

Four lists were created, such that the three conditions were counterbalanced per verb. In total, these crossed conditions and the controls resulted in four counterbalanced lists of 10 experimental trials and 2 control trials in each list, such that each of the ~ 40 participants in each list saw a given verb only twice, regardless of which of the 12 conditions it was in (Animacy x Se x Context). Trials from Experiments 1a and 1b were randomized, so each participant responded to 24 trials in total. This low number stems from the limited number of verbs we could use from the language, on the one hand, and the constraints of a between-participants design in which we did not want a participant to see the same verb twice, on the other hand.

ANALYSIS. For outlier removal, responses were z-transformed into a continuous variable. Participants were removed from analysis if their responses on the gold standard items are, on average, more than 2 SDs away from the mean ratings across all participants.

Raw ratings on the Likert scale (not z-transformed) were fed into a Bayesian ordinal model (White et al. 2018, Veríssimo 2021) implemented in the R package *brms* (Bürkner 2017) using the `cumulative()` family in `cmdstanr`. Animacy, Se and Context were included as population-level effects (‘predictors’), with Animacy as a random slope by subject and item (group-level or ‘random’ effects). While the choice of priors for Bayesian models can be the subject of its own analysis, two things to keep in mind are the value of prior/posterior predictive checks (Nicenboim et al. 2023) and that the *brms* default priors often provide good starting points, as does any weakly informative prior compared to a uniform prior (e.g. Veríssimo 2021). Since we did not have previous hypotheses or results to draw on, we chose to use the default priors (Stan’s ‘improper flat’ prior for predictor means; half-Student t-distribution with 3 degrees of freedom and scale factor 2.5

for random intercepts and predictor standard deviations; all correlation matrices equally likely for calculation of random slopes). The results of the predictive checks can be found in the OSF script.

As noted in the main text, ANIMACY and SE were sum coded. CONTEXT was treatment coded with Neutral as the baseline level (the intercept). The model outputs are given in tables 6–7.

<INSERT TABLE 6 ABOUT HERE>

<INSERT TABLE 7 ABOUT HERE>

Given the small number of observations per condition, the models vary in how well they can estimate group-level (‘random’) effects. The brms output shows correlations between each group-level effect and the slope for Animacy to be, in Experiment 1a, 0.73 for Verb and 0.04 for Participant; and in Experiment 1b, 0.70 for verb and 0.54 for Participant.

These models can also generate predicted ratings. The OSF repository contains code and figures which model two-way and three-way interactions, showing how likely the model would find a specific rating on the Likert scale for each conditions, for example how likely a ‘7’ rating is.

A.2. EXPERIMENT 2.

PARTICIPANTS. Recruitment followed the same procedure as for Experiment 1, resulting in $N = 33$ (40 before exclusions).

PROCEDURE. Participants were presented with two sentences, lying on opposite sides of an unlabeled 7-point Likert scale. They were asked which of the sentences ascribes greater responsibility to the subject.

Materials were presented visually using PCIBex (Drummond 2007, Zehr & Schwarz 2018). Two practice trials preceded the main experiment, in which the order of trials was randomized.

MATERIALS. Verbs in Experiment 2 were either internally caused, limited-control or in-control, although this difference was not coded as a condition. Four verbs were sampled from each verb class. All subjects for these verbs were inanimate. The contrast of interest was between the *se*-marked form and the unmarked form. All contexts were ‘neutral’, in the terminology of Experiment 1:

- (57) *Quelle forme attribue le plus de responsabilité au sabre dans le procès?*
 ‘Which form attributes more responsibility to the sabre in the process?’
- a. Le sabre a rouillé.
 the sabre has rusted
 ‘The sabre rusted.’
 - b. Le sabre s’est rouillé.
 the sabre SE is rusted
 ‘The sabre rusted.’

Control items were created by using four alternating causatives (with a choice between a causative statement and the corresponding unmarked anticausative statement), two naturally reflexive verbs and two naturally disjoint verbs (with a choice between a *se*-passive statement and the corresponding periphrastic passive statement). We expected the causative statement and the *se*-passive statement to be judged as assigning more responsibility to the theme (the latter because the *se*-marked form was the only one yielding a semantically reflexive reading, which was pragmatically odd but nevertheless possible).

Since verbs did not repeat from trial to trial, each participant saw all $3 \times 4 = 12$ critical items, as well as 8 control items.

ANALYSIS. The analysis followed the same procedure as in Experiment 1, except that the choice on the likert scale was converted to a preference between 1 for the unmarked variant and 7 for the marked variant.

Since there was no manipulation between conditions, the regression consisted of a population-level (‘fixed’) intercept and two group-level (‘random’) intercepts. Results are given in Table 8.

<INSERT TABLE 8 ABOUT HERE>

<INSERT FIGURE 8 ABOUT HERE>

B. APPENDIX 2: CORPUS DATA. This appendix contains natural occurrences of grammatical but suboptimal construals with limited-control and in-control verbs used as anticausatives. Recall that with human DPs, we expect $\pm se$ limited-control verbs to generally form anticausatives without *se*, and $\pm se$ in-control verbs to generally form anticausatives with *se*. Although suboptimal, these forms are nevertheless part of the grammar and should therefore be occasionally found in corpora. We list some natural

examples of these unexpected forms below (there were found on the internet or via Sketchengine).

(58) *limited-control ±se anticausatives with se*

- a. Arff, merci; Teret (**me rougi**[s] d'être comparé à Léonardo (Da arff, thanks teret me blush.PRST.1SG to.be compared with Léonardo da Vinci, pas l'autre))
Vinci not the other
'Arff thanks, I'm blushing to be compared to Leonardo (da Vinci, not the other one).'
- forum.millesaisons.fr/forum/viewtopic.php?t=3624&start=20
- b. Arrête tu va[s] me faire **me rougir**, mon cœur. Il[s] m'appartiennent tous!
stop you will me make me blush my heart they me belong all
Et toi avec!
and you with
'Stop, you'll make me blush, my sweetheart. They all belong to me, including yourself!'
lemondededuralas.org/t57p75-demande-de-classe-de-changement-de-classe
- c. Deux magnifiques jumelles, une, aux cheveux d'argents et l'autre
two magnificent twins one with hair of silver and the other
au[x] cheveux plutôt rosâtre[s]. Le jeune homme ne put s'empêch[er]
with the hair rather pinkish the young man NEG could SE help
de **se rougir**, face [à] cette vision paradisiaque.
to SE blush front to this vision heavenly
'Two magnificent twins, one with silver hair, the other one with rather pinkish hair. The young man couldn't help blushing in front of this heavenly vision.'
orkna.forumsrpg.com/t255-pervers-ou-pas-alexander-rina
- d. Je **me rougis** encore en pensant à un moment où j'ai utilisé un
I SE redden still while thinking at a moment where I have used a
slur et après coup j'ai réalisé que c'était inapproprié.
slur and after the fact I have understood that it was inappropriate
'I'm still blushing while thinking at a moment where I used a slur and realized after the fact that it was inappropriate.'
x.com/sinusoid_wave/status/1232733463415054336?s=20

- e. Dans mon cas, il n'y a pas eu de perte de poids, parce que je n'avais
 in my case there hasn't been of lose of weight because I NEG had
 pas besoin, mais je **me suis rajeuni**, ma peau a retrouvé beaucoup
 NEG need but I SE got younger my skin has found back a lot of
 d'éclat.
 brilliance

'In my case, I didn't lose any weight, because I didn't need to, but I looked
 younger and my skin was much more radiant.'

www.calculersonimc.fr/189-des-proteines-et-du-soja-contre-lobesite/

(59) *in-control ±se anticausatives without se*

- a. Moi aussi **j'ai plié** de rire lors du montage aussi.
 me too I have bent from laugh during the editing
 'Me too I bent over in laughter during the editing too.'
www.rc-decouverte.com/t181-nouveau-du-var-scaleou-83
- b. Bojack apparu[t] genou en avant dans le thorax de Trunks, qui **plia**
 Bojack appeared knee forward in the thorax of Trunks who bent over
 sous la douleur.
 under the pain
 'Bojack appeared knee first in Trunks' chest, who buckled under the pain.'
www.lunionsacre.net/viewtopic.php?f=42&t=7116&p=279441
- c. il ne cessait de le prouver depuis qu'il **avait approché** d'elle.
 he NEG stopped to it prove since that he had approached of her
 'He didn't stop proving it since he approached her'.
www.letempsdunrp.com/t4654p10-adrielle-beneath-your-beautiful
- d. On dirait qu'il **a ramolli** à leur contact. C'était le seul
 one WOULD SAY that he has melted at their contact. He was the single
 normal.
 normal
 'One would say he melted at their contact. He was the single normal guy.'
www.psychiatries.be/reconnaissance-et-emancipation

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NOTES

¹The restrictions on the distribution of *se* illustrated in 1b only hold for the anticausative use of these verbs. Since verbs undergoing the causative alternation have a transitive variant, and since reflexivization can apply to all transitive predicates, all these verbs can undergo literal reflexivization. Thus, *changer* in 1b can in fact combine with *se* in the context of a single DP, but the verb is then part of a semantically reflexive construal. Semantically reflexive construals can be distinguished from *se*-marked anticausatives in various semantic ways, for example, in that only the former allow modification by an intensifier, as in (i) (e.g. Schäfer & Vivanco 2016).

- (i) Celui qui se change lui-même change le monde.
 the one who SE changes himself changes the world
 ‘The one who changes themselves changes the world.’ (instagram.com)

²On this point, we thank Hans Kamp for discussion.

³Control is independent from desire and foreknowledge, which are two other dimensions involved in the concept intentionality for Egré 2014 and in the concept of agency for Martin 2023, Joo et al. 2024. For instance, humans typically exert control on their changes in position or posture, but such changes can be performed while the agent does not know that their action can be described with the VP. As an example, I can get closer to a location and control my movements while doing so without knowing that I’m getting closer to this location (because I ignore its existence, for instance).

⁴In typological studies on grammatical voices, verbs like *move* (in their intransitive use) are sometimes considered to instantiate two different voices depending on whether they combine with animate or inanimate subjects. For instance, Geniušienė 1987: 260 considers such verbs to be anticausatives with non-human subjects, and AUTOcausatives with human subjects, where the (single) human DP is both an agent and an undergoer, ‘causing [their] own body to move’ (p. 37). On this point, we follow Bahrt 2021: 37 in the view that the animacy-related distinction between anti- and auto-causatives is spurious. Verbs like *move* instantiate the very same anticausative semantics both with human and non-human subjects (i.e. the single DP is a mere undergoer in the syntax/semantics; when these verbs are used as anticausatives, the agentive control that human undergoers typically enjoy on their change is not encoded in the formal grammar and only reflects conceptual knowledge about in-control events).

An interesting question is whether in-control verbs like *move* in English can also have a semantically reflexive reading so as to express the agent in the syntax/semantics, despite the absence of a reflexive marker, and if so, how this reading is generated in the grammar. One way to proceed is to stipulate a covert reflexivizer, as for instance Jo 2019 does for grooming verbs like *wash/dress* in English. A second way is to resort to middle Voice (Spathas et al. 2015), existentially binding the variable associated with the role agent, and to analyse the inference that the undergoer (expressed by the DP) is also the agent as a pragmatic one. While such solutions are plausible for English, they are not for *–se* ACs like *bouger* ‘move’ in French, since reflexive semantics must be overtly expressed by the reflexive marker *se* in French (Kayne 1975, Reinhart & Sioni 2004). In other words, while English *move* might perhaps be ambiguous between an anticausative and a semantically reflexive reading in English, French *bouger* is unambiguously anticausative (in its intransitive use).

⁵We thank John Beavers for pushing us to emphasize this point.

⁶We remain noncommittal in this paper whether the relevant contrast is between humans and non-humans or animates and inanimates; the strongest intuitions concern humans, but there could well be a cline of relevant animacy, with animals or even artefacts patterning more with humans in some contexts than in others.

⁷One further verbal diathesis formed with *se* in French and other Romance languages is the *se*-passive (or medio-passives). We believe that *se*-passives are sometimes involved in competition-based effects similar to those identified with *se*-marked anticausatives, but we leave this topic for another occasion.

⁸ We leave aside the interesting question of whether the French data under discussion can also be accounted for adopting an analysis of marked anticausatives as semantically reflexive, discussed in Chierchia 2004, Koontz-Garboden 2009 and Lundquist et al. 2016. A detailed comparison of these two analyses is beyond the scope of this paper, but see Horvath & Sioni 2011, 2013, Beavers & Koontz-Garboden 2013a, b and Schäfer & Vivanco 2016 for relevant discussion.

⁹French semantically reflexive verbs are analyzed as transitive in Sportiche 2014 and Alexiadou et al. 2015, and as intransitive in, for instance, Reinhart & Sioni 2004; see also Labelle 2008 for arguments against generating *se* in object position. But all analyses converge in the view that reflexive construals assign an agentive role to the DP, which is the most relevant point here.

¹⁰For Reinhart 2002, Reinhart & Siloni 2005, *se* is a meaningless case absorber in anticausatives (and beyond), for Schäfer 2008, Alexiadou et al. 2015, *se* is an expletive (i.e. meaningless) external argument in anticausatives.

¹¹For the phenomenon we look at, search for perspicuity plays the central role. The free variation between forms with and without *se* observed with $\pm se$ -ACs in the context of a non-human subject (and confirmed in Experiments 1a-b presented in section 3) shows that economy constraints (and the related submaxim of Manner BE BRIEF) do not seem to play a role in the distribution of *se* with $\pm se$ verbs, as it should prohibit $\pm se$ optionality without further qualifications. Economy does not account either for the distribution of $\pm se$ forms in the context of human DPs, since the clearest/most perspicuous form is sometimes the shortest one, without *se* (with limited-control $\pm se$ verbs), sometimes the longest one, with *se* (with in-controls verbs).

Many pragmatic studies resorting to the maxim of Manner have concentrated on BE BRIEF and the related AVOID OBSCURITY OF EXPRESSION submaxims, a common assumption being that the choice for a more prolix form over a simpler form conveying the same truth-conditions signals the intention to convey a marked meaning (Horn 1989, Levinson 2000, Rett 2015, 2020 among others). The phenomenon we are looking at here is different in nature, as the morphologically marked form (with *se*) does not systematically signal a marked meaning compared to the morphologically unmarked form (without *se*).

¹²Degen & Tanenhaus 2015 ultimately aim to explain why the vague indefinite *some* triggers delays in response time when a cardinal indefinite such as *two* is a salient alternative, as in experiments by Huang & Snedeker 2011 they are discussing, challenging the view that these delays result from the costly computation of the quantity implicature from *some* to *not all*.

¹³The motivation of this interference, in Gricean terms, comes from the maxim of Manner: if there is a less ambiguous quantifier (e.g. *two*) that could have been chosen to refer to a particular set size, the speaker should have used it. If she didn't, it must be because she meant something else.' (Degen 2013: 68)

¹⁴For the case of limited-control $\pm se$ AC verbs, it may be that the conventionalization process of the 'agentive undergoer inference' triggered by the form with *se* is more advanced for some verbs than others, making the suboptimal form more marked for some verbs than for others.

¹⁵Note that in sentences like 5a, the causer-PP makes clear that the human undergoer is

not the intentional agent of the change. Therefore, the DP's referent cannot be associated with 'strong' agentive properties such as those presupposed by agent-adverbials like *cautiously* or *skillfully*. This is why in the presence of such agent-oriented adverbials, a causer-PP is not felicitous anymore, as shown in the example below:

- (i) Jeanne s'est (#élégamment/ #prudemment/ #doucement) pliée en deux de
 Jeanne SE is elegantly cautiously gently bent in two from
 douleur.
 pain
 'Jeanne elegantly/ cautiously/ gently bent over in pain.'

Still, in such contexts where the causer-PP enhances the anticausative construal, the human DP can nevertheless be conceived as an agent in a weak sense, under which agency is reduced to the property of effectivity (in this weak sense of 'agent', Jeanne just needs to be the doer/effector of her bodily movements). (We refer to Martin 2023 and Joo et al. 2024 for an analysis of the role of agent as ambiguous between a weak, 'effector' meaning, and a stronger meaning involving more agentive dimensions than effectivity, such as agent control.)

Such sentences with the form with *se* of in-control verbs are thus potentially ambiguous. But the anticausative reading is preferred; for instance, *Il s'est plié de douleur* 'He bent over in pain' is more natural after a *What happened to him?* question than after a *What did he do?* question, but the latter is also possible (see also the discussion on examples 36 in section 3).

In contrast, the *–se*-variant of limited-control verbs being unambiguously anticausative, it does not constitute a felicitous answer to a *What did they do?* question (see *Qu'est-ce qu'il a fait? #Il a pâli/rougi*. 'What did he do? He turned pale/blushed.').

¹⁶ Indeed, while anticausatives remove the agent from the syntax/semantics and thereby de-emphasize this role, they can be used even if the context makes clear that the change was triggered by an agent. As Rappaport Hovav (2014: 23) notes after McCawley (1978) for instance, anticausatives are perfectly acceptable in a context making clear that an agent causes the reported change, see (ia). One can similarly observe that unaccusative verbs with a human DP are acceptable in a context making clear that the human undergoer is also the agent of the change; see for instance (ib).

- (i) a. I pushed and pushed on the door, and finally it opened. (McCawley 1978, cited in Rappaport Hovav 2014: 23)

- b. John squeezed the trigger, and thereby died.

The case of $\pm se$ in-control verbs is different, precisely because the choice of the $-se$ -variant is interpreted as signalling the lack of agency of the DP's referent.

¹⁷Jackendoff 1972: 44 argues that on its stative use, English *touch* associates the roles Theme and Goal/Location to its arguments.

¹⁸According to what we call the 'Aspectual Claim', ACs marked with *se* focus on the achievement of the result state, while ACs without *se* focus on the process of the verbal event (Labelle 1992, Labelle & Doron 2010). In a distinct variant of the Aspectual Claim, Legendre et al. 2016 and Legendre & Smolensky 2017 propose that only with $\pm se$ AC-verbs, that is, if a choice is available, the marked variant necessarily carries a 'completion interpretation', while the variant without *se* necessarily carries a 'partial completion interpretation'. For them, this amounts to saying that the former are interpreted as telic, and the latter as atelic predicates. However, differently from Labelle 1992 and Labelle & Doron 2010, these authors explicitly assume that no such specialization in meaning holds for $-se$ AC-verbs and $+se$ AC-verbs. They analyse their specific version of the Aspectual Claim within a bi-directional optimality theoretic system that involves blocking and antiblocking of particular meaning-form pairs. Martin & Schäfer 2014 showed on the basis of attested examples found in corpora that the Aspectual Claim is based on faulty generalizations. As they show, AC-verbs with and without *se* do not consistently differ in terms of completion entailment.

¹⁹Later work has argued that the distinction between internal and external causation is empirically and conceptually problematic, and grammatically irrelevant (see in particular Alexiadou 2014 and Rappaport Hovav 2014, 2020). An obvious problem is the circularity in the argumentation: 'verbs are classified in an intuitive way and then when the data go contrary to the classification, verbs are suggested to be either wrongly classified or to allow more than one classification' (Rappaport Hovav 2020: 227). Relatedly, in some languages (including English), verbs typically classified as internally caused like *wilt* can be used transitively with an external causer subject, and sometimes even with an agentive subject (Wright 2002). For Rappaport Hovav 2020: 245, the reason why internally caused change-of-state verbs are most of the time used intransitively is not grammatical, but rather conceptual: external causal factors for the changes expressed by these verbs are just very expected to occur, which is why they remain unnamed.

²⁰We therefore do not put in our class of limited-control verbs so-called

internally-caused change-of-state $\pm se$ verbs such as *flétrir* ‘wilt’ or *rouiller* ‘rust’.

The latter verbs only combine with a human subject if its referent is metaphorically reinterpreted as a (non-agentive) vegetal or mineral entity (as in e.g. *je (me) flétris* ‘I’m wilting’ or *je (me) rouille* ‘I’m rusting’). Verbs like *sleep* or *hiccup* also take a human subject and also express events that cannot be controlled, but these events are activities, not changes. These verbs are thus limited-control activity (intransitive) verbs. We are not concerned with these verbs here.

²¹All these verbs alternate in French, but $-se$ verbs tend to be more often used in intransitive construals, while $+se$ verbs tend to be more often used in transitive construals (Heidinger 2010, 2015).

²²Reflexive semantics is morphologically or lexically marked across languages (e.g. Reinhart & Reuland 1993, Kastner 2017).

²³The distinction is equally irrelevant for ACs with a fixed morphological behavior. With such verbs, non-human subjects are unproblematic in an internally caused as well as an externally caused setting, as illustrated with the $+se$ AC *se briser* ‘break’ in ia-b and with the $-se$ AC *exploser* ‘explode’ in (iia-b).

- (i) a. le téléphone construit par Huawei rencontrerait (...) de gros soucis
 the telephone built by Huawei meet.COND.3SG of big problems
 de fragilité au niveau de ses vitres qui se briseraient toutes seules
 of fragility at.the level of its glasses which SE break all alone
 selon de nombreux utilisateurs.
 according to of a lot of users
 ‘The phone built by Huawei has many problems of fragility with regard to its
 panes which break by themselves according to many users.’
 (internal cause, Frtenten20, beegreek.fr)
- b. la majorité des noyaux se brisent sous l’action des photons.
 the majority of.the kernel SE break under the.action of.the photons
 ‘the majority of kernels break under the action of photons.’
 (external cause, Frtenten20, astrosurf.com)

- (ii) a. L'Iphone a vraiment explosé de lui-même.
 the.Iphone has really exploded by itself
 'The iPhone really exploded by itself.' (internal cause, Frtnten20, iphon.fr)
- b. Certaines vitres explosent sous l'action du vent.
 some glasses explode under the.action of.the wind
 'Some glasses explode under the action of the wind.'
 (external cause, Frtnten20, keraunos.org)

²⁴We used a default prior, Stan's 'improper flat' prior for predictor means, as we did not have previous results to draw on. See Appendix 1 for additional details.

²⁵Our post-hoc account for this pattern is that the overt inchoative context helps the interpreter to understand that they should NOT conclude from the speaker's choice of the *se*-variant that they were after the (agentive) reflexive parse, which is precisely according to our analysis the confusing inference typically drawn on the basis of the maxim of Manner (see section 2).

²⁶Some of the verbs listed under 35c can be used *se*-marked with a single human argument, see (ia) below. However, they must remain unmarked when used with a single non-human argument (see (ib)), which we take to indicate that the *se*-marked variant with a human DP is always semantically reflexive, and never anticausative.

- (i) a. Pierre (se) bouge/ (se) recule.
 Pierre SE moves/ SE stepsbackwards
- b. La pierre (#se) bouge/ le ballon (#se) recule.
 the stone SE moves/ the ball SE steps backwards

²⁷We choose to probe the intuition of participants on the dimension of responsibility rather than agency, since graded responsibility attribution is more frequent in the layman language than graded agency attribution, and relatedly, it is more usual to attribute overtly responsibility rather than agency to non-human entities in ordinary language (e.g. *Which sentence assigns more agency to the chair/the car in the event* is a less natural question than our test question in usual language).

²⁸Thus the lexical semantic subclasses of $\pm se$ ACs discussed in the previous sections do not matter here, because they can be contrasted only via the assumptions we hold

by default about human participants in the events respectively denoted by verbs of each subclass.

²⁹Items with a non-human subject in the reflexive context were put in the present tense rather than the *passé composé*, for the latter tense/aspect morphology would trigger an anomaly in this context independently of whether the reflexive morphology is present or not. Since our goal was to test how the presence vs. absence of reflexive marker affects the acceptability of the sentence, we built the examples so as to make them as natural as possible independently of this factor.

³⁰A reviewer suggests that examples such as 47 may receive an analysis as a *se*-passive with the implicit agent being identified with the possessor of the internal argument DP (e.g. the insect in 47). In this perspective, examples such as 47 are derived from an active string such as (i) below (see Lundquist 2016: 184-185 for an analysis of similar examples involving the syncretic morpheme *-s* in Swedish).

- (i) Dans cette situation, l'insecte noircit sa carapace pour échapper aux prédateurs.
 'In this situation, the insect blackens its carapace in order to escape the predators.'

While this analysis has some appeal, we see a number of reasons speaking against it. First, it is not entirely clear to us how a sentence such as 47, where the possessor of the internal argument is a referential expression, could technically be derived as a passive of (i), where the possessor of the internal argument is a possessive pronoun. Second, it is not clear how the implicit external argument could be co-valued with the possessor of the internal argument, the latter being an R-expression. Third, there is a difference in meaning between 47 and (i), namely that 47 does not agentivize the insect the way (i) definitely does. We take this to indicate that the *se*-variant of examples such as 47 well and truly has an anticausative reading (47 also has, in principle, a semantically reflexive reading, which can be enforced by adding the intensifier *elle-même* ('itself'), the latter being bound by the possessee DP *carapace*).

In general, the passive reading is not a viable option for the *se*-marked test items across our experiments, for they contain no contextual element hinting at the involvement of an implicit agent, which is for Zribi-Hertz 1982 a condition for the *se*-passive to arise. The single exception is one of the 8 items with non-human subject in the reflexive context, namely example 48, where the implicit agent (understood as the user of the charger) controls into the purpose clause (this example, however, can also have an anticausative reading). 48 is also the single item where the *se*-variant can be felicitously

replaced with a standard *be*-passive. In the other examples including 47, a paraphrase with a *be*-passive is odd, as it suggests the involvement of an agent different from the possessor in the DP (due to the well-known disjoint reference effect holding between the implicit external argument of passives and any R-expression inside the VP; see e.g. Bhatt & Pancheva 2017, Schäfer et al. 2024). For instance, *#La carapace de l’insecte est noircie pour échapper aux prédateurs* ‘The carapace of the insect is blackened in order to escape the predators’ is quite odd in French, just like its English counterpart. That being said, even if some participants accessed a passive reading for some items of the +*se*.Non-human.ReflexiveContext condition, this would not affect our general point that the preference we observe for the *se*-marked variant with a human DP in the reflexive context does not hold with a non-human DP.

³¹The findings went in the predicted direction in 73.8% of trials in Experiment 1a and 77.5% of trials in Experiment 1b, both $p < 0.0001$ in sign tests.

	Human argument	Non-human argument
$\pm se$ limited-control verbs (<i>se</i>) <i>rougir</i> ‘redden/blush’	variant without <i>se</i> preferred (<i>Experiment 1a</i>)	no preference between variants (<i>Experiment 1a</i>)
$\pm se$ in-control verbs (<i>se</i>) <i>plier</i> ‘bend, fold’	variant with <i>se</i> preferred (<i>Experiment 1b</i>)	no preference between variants (<i>Experiment 1b</i>)
All $\pm se$ verbs		variant with <i>se</i> preferred to convey responsibility of Non-human (<i>Experiment 2</i>)
All $+se$ verbs (including limited-control) <i>s'affaiblir</i> ‘become weak(er)’	no choice	no choice
All $-se$ verbs (including in-control) <i>changer de position</i> ‘change in position’	no choice	no choice

TABLE 1. Claims in the current paper on French *se* across AC verbs and related experiments.

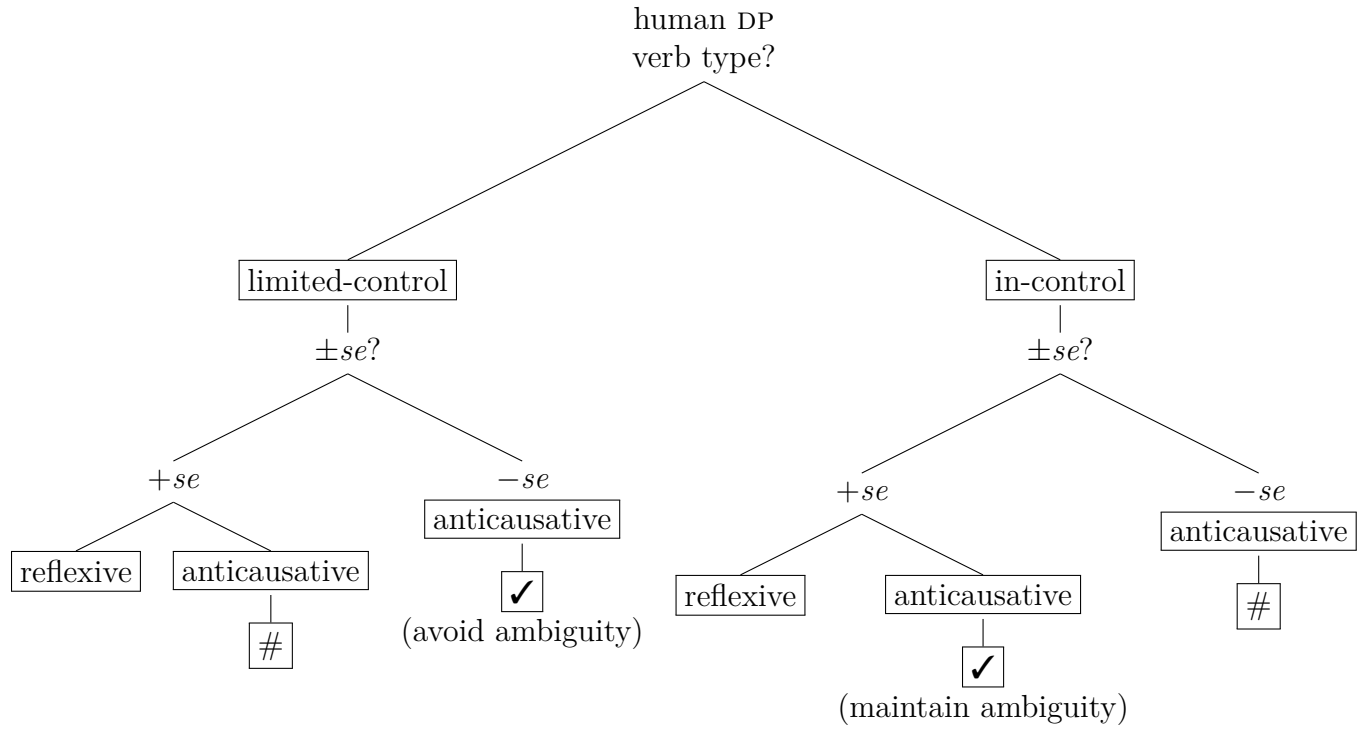


FIGURE 1. Pragmatic reasoning and resulting preferences with human subjects and $\pm$ ACs ('reflexive' = semantically reflexive reading, 'anticausative' = anticausative reading).

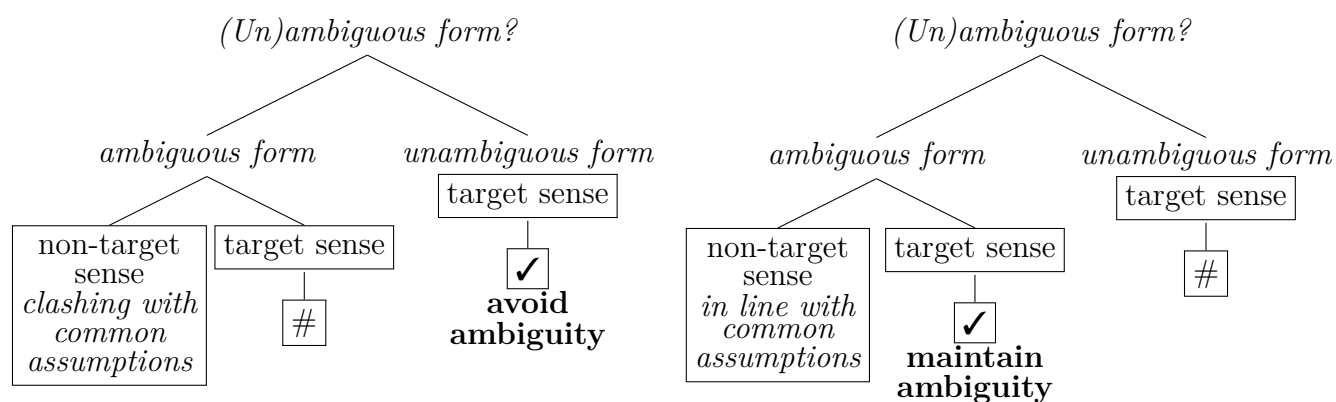


FIGURE 2. Two optimal ways to manage ambiguity.

Est-ce que la phrase suivante est naturelle?

Djamila a pâli à l'annonce de l'infidélité de son amoureux.

Tout à fait naturelle ○ ○ ○ ○ ○ ○ **Pas du tout naturelle**

FIGURE 3. Example of stimuli of Experiment 1a rating scale task.

Context	Animacy	+se M	SE	-se M	SE
Inchoative	Human	4.109	± 0.197	6.321	± 0.110
	Non-human	5.167	± 0.190	5.583	± 0.173
Neutral	Human	3.218	± 0.203	6.526	± 0.109
	Non-human	4.616	± 0.200	5.2821	± 0.183
Reflexive	Human	4.904	± 0.189	3.551	± 0.198
	Non-human	5.917	± 0.156	6.449	± 0.112

TABLE 2. Raw means (M) and standard errors (SE) for Experiment 1a.

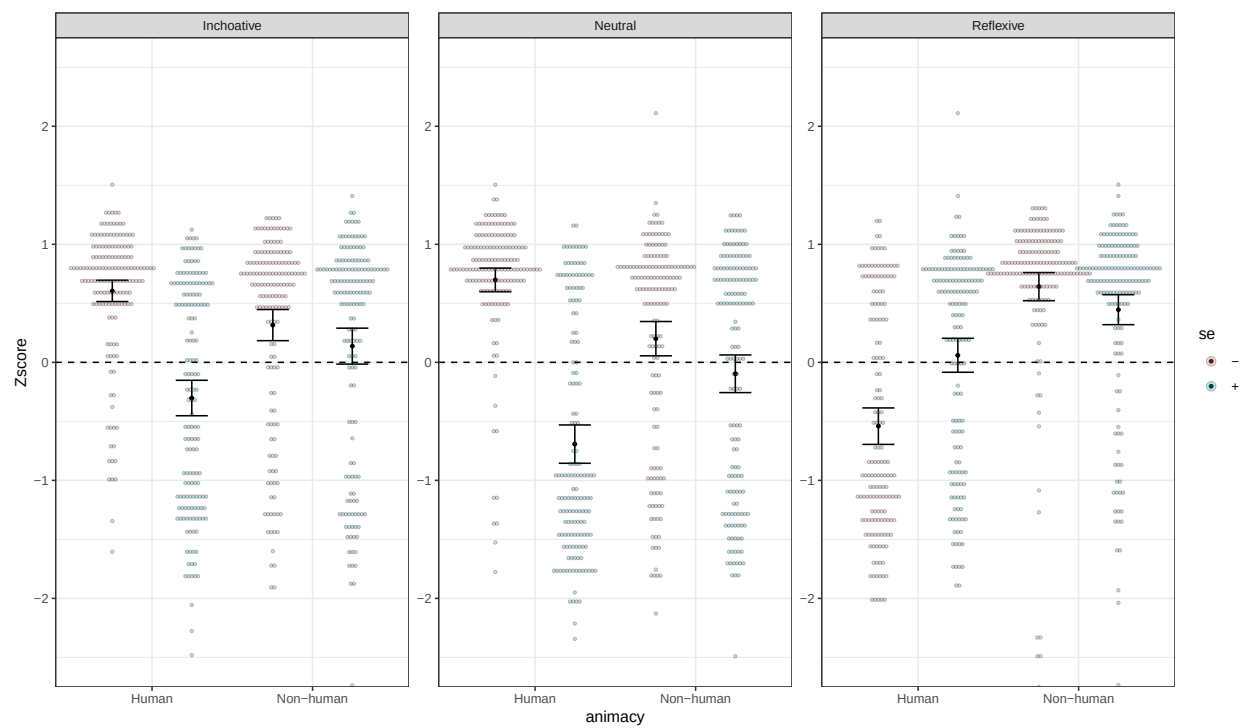


FIGURE 4. Results of Experiment 1a (limited-control verbs).

	Estimate	Est. Error	95% CI
AnimacyHuman	1.66	0.79	[0.12,3.22]
Se	-0.78	0.23	[-1.25,-0.35]
ContextInchoative	0.46	0.24	[-0.01,0.92]
ContextReflexive	1.11	0.36	[0.42,1.81]
AnimacyHuman:Se	-2.87	0.36	[-3.58,-2.15]
AnimacyHuman:ContextInchoative	-1.28	0.38	[-1.99,-0.56]
AnimacyHuman:ContextReflexive	-4.50	0.46	[-5.43,-3.61]
Se:ContextReflexive	-0.10	0.36	[-0.79,0.60]
AnimacyHuman:Se:ContextInchoative	1.35	0.49	[0.37,2.30]
AnimacyHuman:Se:ContextReflexive	4.82	0.51	[3.84,5.81]

TABLE 3. Relevant predictors from the Bayesian ordinal model, Experiment 1a (limited-control).

Context	Animacy	+se M	SE	-se M	SE
Inchoative	Human	5.590	± 0.167	3.506	± 0.201
	Non-human	6.051	± 0.147	6.237	± 0.124
Neutral	Human	5.904	± 0.154	3.628	± 0.200
	Non-human	5.641	± 0.172	5.269	± 0.192
Reflexive	Human	5.891	± 0.153	2.994	± 0.196
	Non-human	6.308	± 0.124	5.654	± 0.165

TABLE 4. Raw means and standard errors for Experiment 1b.

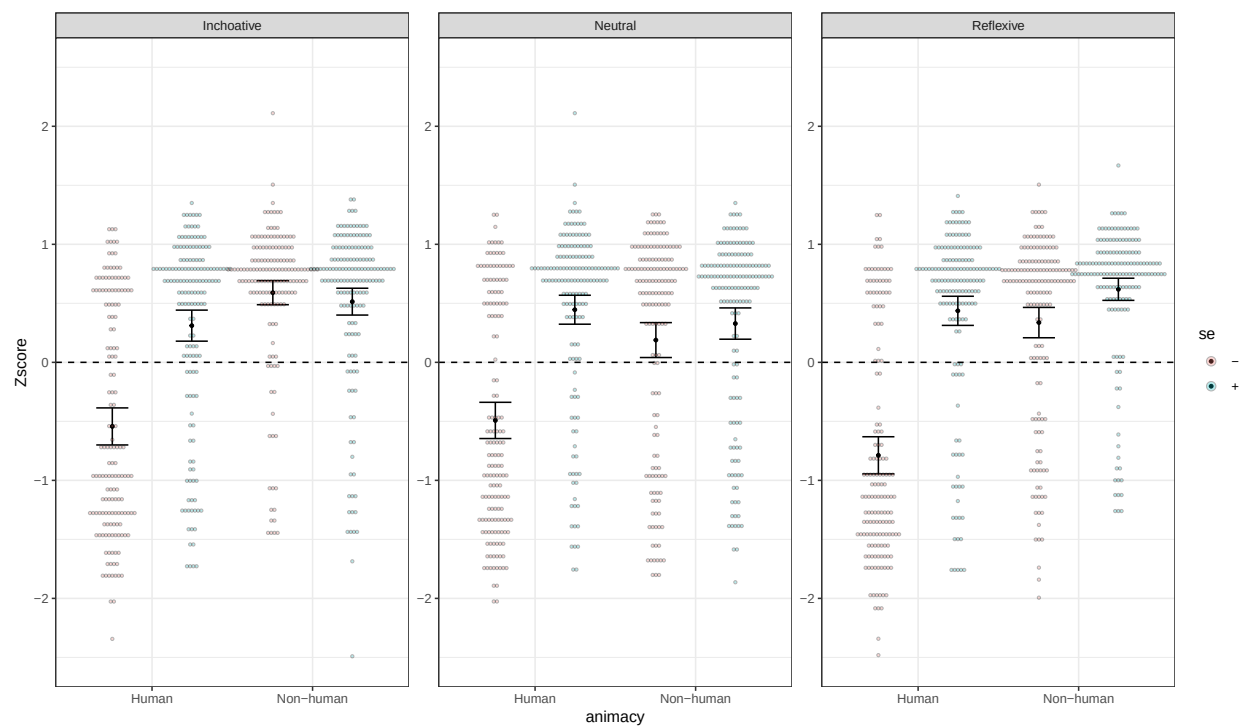


FIGURE 5. Results of Experiment 1b (in-control verbs).

	Estimate	Est. Error	95% CI
AnimacyHuman	-1.32	0.68	[-2.63,0.09]
Se	0.38	0.23	[-0.07,0.85]
ContextInchoative	0.87	0.25	[0.39,1.39]
ContextReflexive	0.22	0.23	[-0.21,0.67]
AnimacyHuman:Se	1.57	0.32	[0.93,2.21]
AnimacyHuman:ContextInchoative	-1.25	0.34	[-1.93,-0.61]
AnimacyHuman:ContextReflexive	-1.06	0.32	[-1.69,-0.45]
Se:ContextReflexive	0.44	0.34	[-0.24,1.09]
AnimacyHuman:Se:ContextInchoative	0.29	0.47	[-0.61,1.22]
AnimacyHuman:Se:ContextReflexive	0.08	0.47	[-0.84,0.99]

TABLE 5. Relevant predictors from the Bayesian ordinal model, Experiment 1b (in-control).

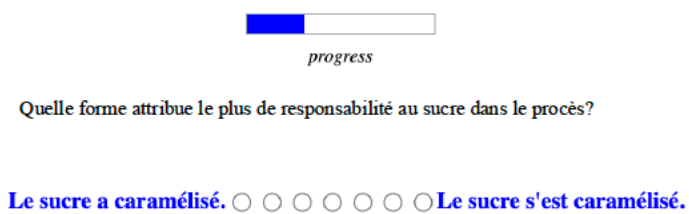


FIGURE 6. Example of stimuli for Experiment 2 responsibility scale task.

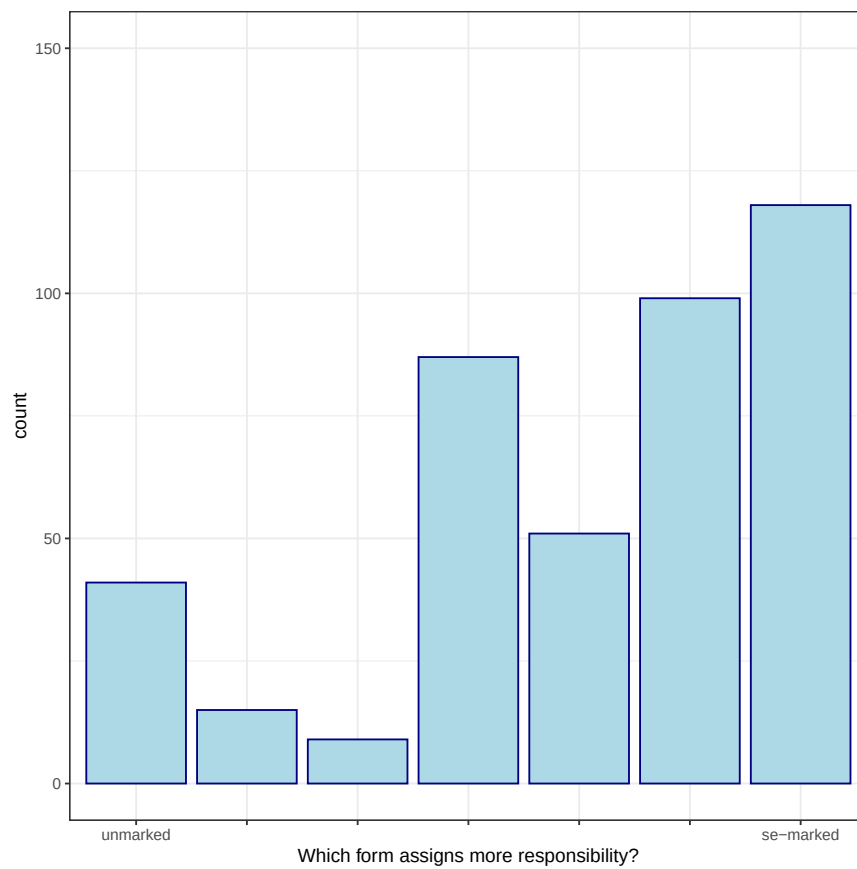


FIGURE 7. Results of Experiment 2 (non-human responsibility, inanimate subjects).

	Estimate	Est. Error	95% CI
Intercept[1]	-2.15	0.56	[-3.18,-0.95]
Intercept[2]	-1.62	0.56	[-2.66,-0.44]
Intercept[3]	-1.30	0.56	[-2.33,-0.12]
Intercept[4]	-1.09	0.56	[-2.13,0.09]
Intercept[5]	-0.67	0.56	[-1.71,0.50]
Intercept[6]	-0.06	0.56	[-1.11,1.13]
AnimacyHuman	1.66	0.79	[0.12,3.22]
Se	-0.78	0.23	[-1.25,-0.35]
ContextInchoative	0.46	0.24	[-0.01,0.92]
ContextReflexive	1.11	0.36	[0.42,1.81]
AnimacyHuman:Se	-2.87	0.36	[-3.58,-2.15]
AnimacyHuman:ContextInchoative	-1.28	0.38	[-1.99,-0.56]
AnimacyHuman:ContextReflexive	-4.50	0.46	[-5.43,-3.61]
Se:ContextInchoative	0.22	0.33	[-0.42,0.88]
Se:ContextReflexive	-0.10	0.36	[-0.79,0.60]
AnimacyHuman:Se:ContextInchoative	1.35	0.49	[0.37,2.30]
AnimacyHuman:Se:ContextReflexive	4.82	0.51	[3.84,5.81]

TABLE 6. Full results of the Bayesian ordinal model, Experiment 1a (limited-control).

	Estimate	Est. Error	95% CI
Intercept[1]	-2.05	0.53	[-3.04,-0.94]
Intercept[2]	-1.59	0.53	[-2.60,-0.49]
Intercept[3]	-1.21	0.53	[-2.21,-0.10]
Intercept[4]	-1.01	0.53	[-2.00,0.09]
Intercept[5]	-0.63	0.53	[-1.62,0.48]
Intercept[6]	-0.12	0.53	[-1.11,0.99]
AnimacyHuman	-1.32	0.68	[-2.63,0.09]
Se	0.38	0.23	[-0.07,0.85]
ContextInchoative	0.87	0.25	[0.39,1.39]
ContextReflexive	0.22	0.23	[-0.21,0.67]
AnimacyHuman:Se	1.57	0.32	[0.93,2.21]
AnimacyHuman:ContextInchoative	-1.25	0.34	[-1.93,-0.61]
AnimacyHuman:ContextReflexive	-1.06	0.32	[-1.69,-0.45]
Se:ContextInchoative	-0.50	0.35	[-1.20,0.17]
Se:ContextReflexive	0.44	0.34	[-0.24,1.09]
AnimacyHuman:Se:ContextInchoative	0.29	0.47	[-0.61,1.22]
AnimacyHuman:Se:ContextReflexive	0.08	0.47	[-0.84,0.99]

TABLE 7. Full results of the Bayesian ordinal model, Experiment 1b (in-control).

	Estimate	Est. Error	95% CI
Intercept[1]	-3.67	0.52	[-4.69,-2.63]
Intercept[2]	-3.03	0.50	[-4.01,-2.02]
Intercept[3]	-2.17	0.49	[-3.68,-1.73]
Intercept[4]	-0.85	0.48	[-1.78,0.09]
Intercept[5]	0.04	0.48	[-0.90,0.97]
Intercept[6]	1.83	0.49	[0.89,2.78]

TABLE 8. Results of Experiment 2, ordinal Bayesian model.

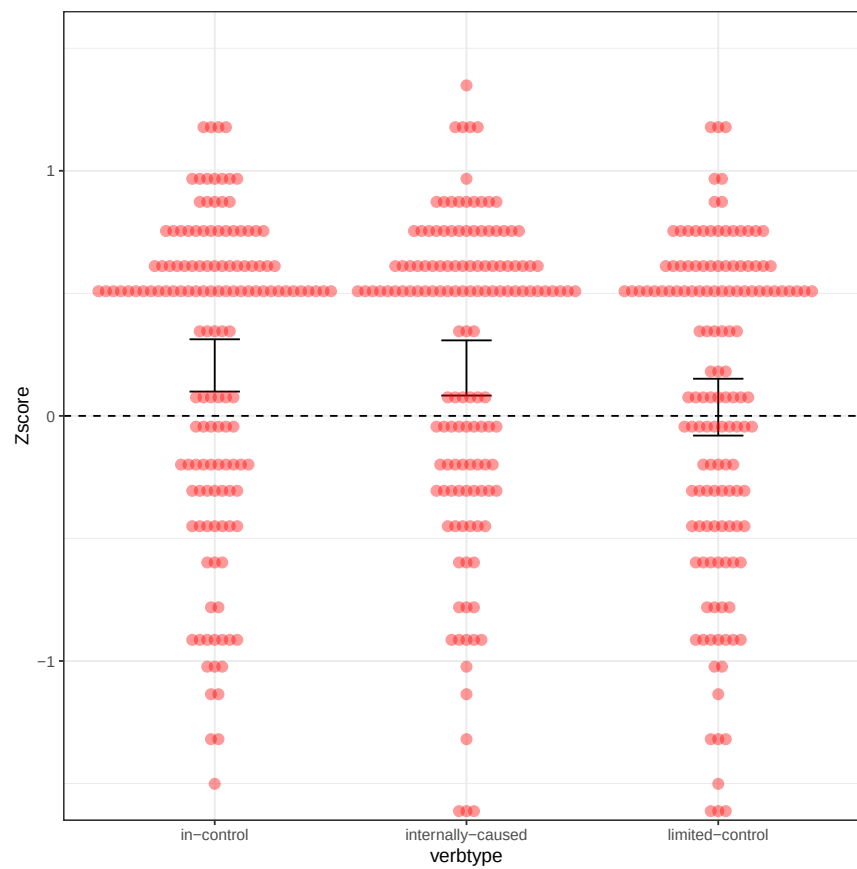


FIGURE 8. Experiment 2: Ratings by Verb Type.